

April 19th, 2026
LOW 4050

$\dot{m}_{fu}(P_c, T, (T_1), V_{fu}) \cdot V_{fu} \cdot \Delta t$
 $(u_{g1} - u_{g1})$
 $\text{Cool prop. Prop. S.I.}$
 $\text{at } t + \Delta t$
 $\left[\begin{array}{l} \rightarrow (u_{g1}, u_{g1}) \text{ at } T_1, S_{c1} \\ \rightarrow (u_{g2}, u_{g2}) \\ \rightarrow (u_{f1}, u_{f1}) \end{array} \right] \quad \chi_2 = \frac{V_2 - V_{f2}}{V_{g2} - V_{f2}}$
 $\dot{m}_{fu}(P_c, T, (T_1), V_{fu}) \cdot V_{fu} \cdot \Delta t - \dot{m}_{ox} h_{f1}(T_1) \cdot \Delta t + u_f(T_1)$
 $-\left[u_{g2}(T_2) + \left(\frac{V_2 - V_{f2}}{V_{g2} - V_{f2}} \right) (u_{g2} - u_{f2}) \right] = 0$
 $h_g(z): B \neq 0$
 $h_g(z) = \frac{0.026}{D_e^{0.5}} \left(\frac{P_c}{P_{c0}} \right)^{0.5} \left(\frac{T_c}{T_1} \right)^{0.5} \left(\frac{A}{A_1} \right)^{0.5} \left(\frac{m}{m_1} \right)^{0.5}$
 $\chi_2 = \frac{S_{c1} - S_{c1}(T_2)}{S_{g1}(T_2) - S_{f1}(T_2)} = f(T_2)$
 $h_c = 0.023 \left(\frac{u_{f1}}{D_e} \right)^{0.8} \left(\frac{P_c}{P_{c0}} \right)^{0.4}$
 $Q = \frac{T_g - T_c}{R_{total}}$
 $\sigma(m) = \frac{T_w}{2E} \left(1 + \frac{9-1}{2} m(z)^2 \right)$
 $A = A_c - V_{k2} \cdot t_{rib}$
 $\lambda_{k2} = \frac{1}{R_{k2}}$



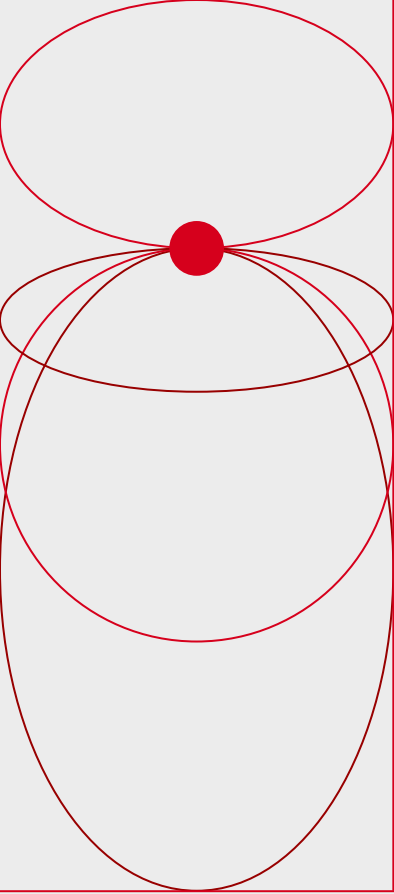
RPI
Experimental
Propulsion Initiative

Agenda

April 2026



- 01** Project Overview
- 02** Concept of Operations & Project Timeline
- 03** Technical Overview
- 04** Subsystem Engineering Review
- 05** Comments and Advice
- 06** Conclusion & Evaluation

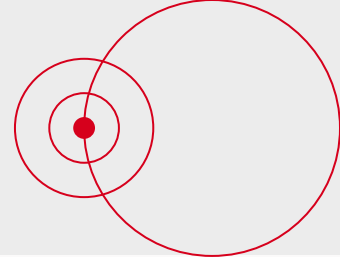


Project Overview

01

Mission Statement

RXPI 2026-2027 Aquila Mission



Design, build, test, and fly a 1500-lbf Nitrous-Ethanol liquid rocket engine (RPU-2 Altair) and flight vehicle (FV-01 Albatross) while training student members in industry-standard aerospace systems engineering practices and maintaining high safety standards.

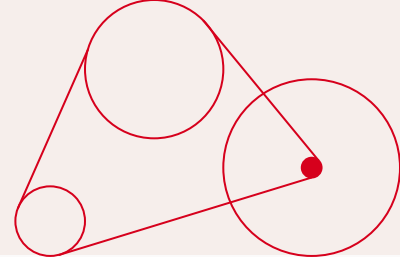
Primary Objectives

- Hot fire RPU-2 at full thrust for target burn duration
- Compete in the Friends of Amateur Rocketry (FAR)
- Ten Cent per Foot Challenge
- Complete a safe, controlled launch at the FAR Launch Site (April-May 2027)
- Demonstrate flight vehicle integration & regenerative cooling capabilities
- Achieve Safe Recovery for Post-Flight Analysis

Secondary Objectives

- Advance team knowledge in liquid propulsion design
- Validate analytical models via hot-fire correlation
- Train members in systems engineering principles such as integration, validation, and responsible engineering.
- Establish reusable infrastructure for future RXPI projects
- Document lessons and engineering design process for our successors

Aquila System Overview



Altair - RPU-2 [RPI Propulsion Unit]

Thrust: 1500 lbf nominal

Propellants: N₂O / Ethanol

Feed System: Pressure-fed
(self-pressurizing N₂O)

Cooling: Regenerative

Ignition: Head-end cartridge igniter

Chamber Pressure: 340 psi

O/F Ratio: 3.2 (central)

Target Isp: ~200 s

Burn Duration: 15–20 s

Albatross - FV-01 [Flight Vehicle]

Diameter: 8 in (concentric tanks)

Length: ~19 ft

Target Apogee: 60,000+ ft AGL

Max Velocity: Mach 1.5–2.0

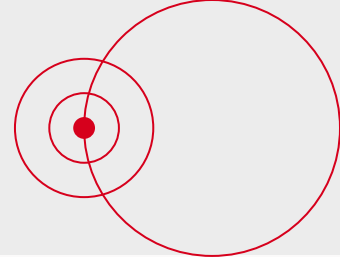
Total Impulse: ~114,000 N·s

Recovery: Dual-deploy parachute

Main Valves: SABV (servo-actuated)

Pressurization: Nitrous
vapor-pressure

Team Organization



Program Directorate

Program Manager
Brian Witanowski

Chief Engineer
Maximilian Slavik

SEI Director
Clara Madison

TOV Director
Roberto Ribay

Subteam Leads

Propulsion
**Reese Mount/
Maximilian Slavik**

Fluids
**Slater
Selinger/Ty Hollis**

**Electronics &
Software**
Sydney Howell

**Airframe &
Recovery**
Riley Dunn

**Ground Support
Equipment**
Jasper Heymann

TECHNICAL REVIEW BOARD (EXTERNAL)

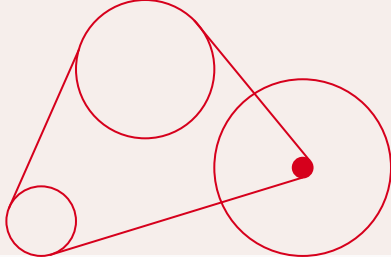
Faculty Advisor / Review Chair: Prof. Ozgur Tumuklu

EHS Representative: [TBD]

Industry SME(s): [TBD]

SRR Action Items – Closure Status

Three action items assigned at SRR (Feb 22, 2026)



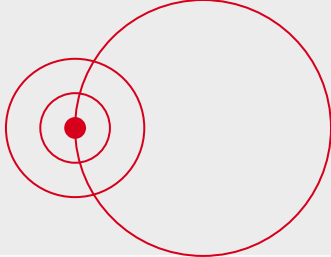
Action Item (SRR)	Owner	Deadline	Status	Evidence / Artifact
Preliminary MOPS & TPMs for key requirements	TOV Proxies, Subsystem Leads	3/15/2026	CLOSED	TPM Sheet
Preliminary verification & validation method per requirement	TOV Proxies, Subsystem Leads	3/15/2026	CLOSED	RVM Sheet V&V Column
Single Point Failure / Fault Tolerance Philosophy in Requirements	SEI, TOV	3/15/2026	CLOSED	FMECA & RVM

Summary

All three SRR action items were completed by the 3/15 deadline; residual work on TPM maturity and SPF requirement flow-down is captured as PDR Entrance 6.2 / 6.9 open items.

PDR Entrance Criteria Status

30 Total Entrance Criteria



29

CLOSED

1

ON TRACK

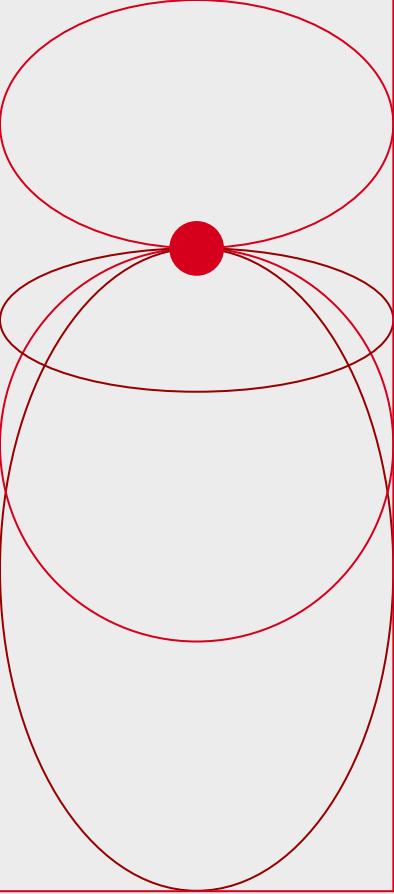
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NOT ON TRACK

Criteria On Track To Close

#	Criterion	Owner	Closure Plan
6.16	Preliminary Disposal Plan (landing & recovery)	A&R	Draft in progress; close by CDR

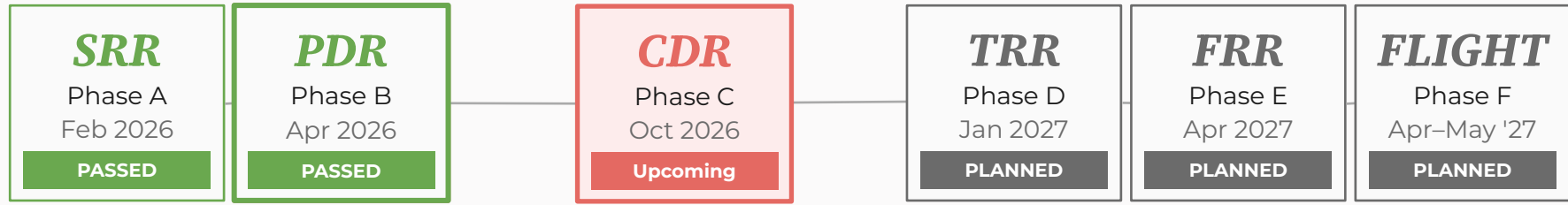
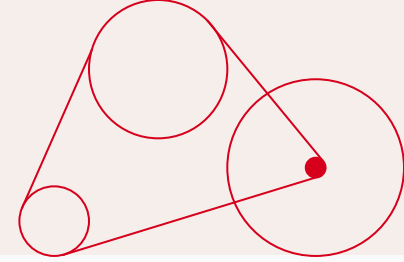
Open item does not block entrance; it has an accepted closure plan



Concept of Operations

Program Manager: Brian Witanowski

V-Model Lifecycle & Schedule Status



Phase B Deliverables – 30% Design

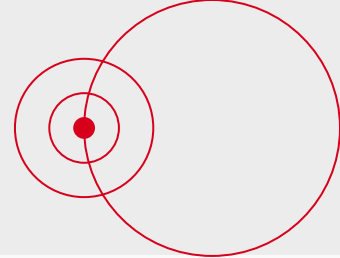
- Trade studies complete for major architecture decisions
- Preliminary Analysis
- Baselined ICDs and preliminary drawing tree
- FMECA completed for major components

Critical Path

- Trade Studies and Further Analysis
- Design Freeze at CDR
- Long-Lead Procurement & Sponsored/External Prints/Manufactured
- Assembly
- Hydrostatic/Cold-Flow in Dr. O's Lab / Ricketts High Bay / '86 Field
- Hot-Fire Test at Weibert's Autoshop
- Vehicle Integration
- Cross Country Transportation & FAR Launch

Budget Overview

Total Aquila Life Cycle Cost Estimate



TOTAL

\$23,390

Baseline estimate

Hardware (valves, sensors, materials)

\$15,490

Propellants (N₂O, ethanol — 2× loads)

\$1,600

Travel to FAR

\$5,000

FAR fees & camping

\$500

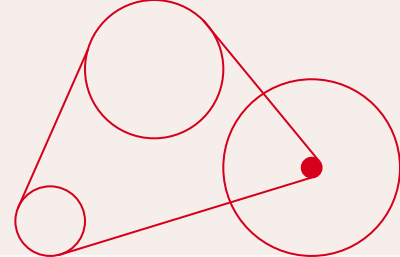
Tools & consumables

\$800

FUNDING STATUS: ~9500 in Monetary and Materials Sponsorships. More sponsors expected after PDR.

Logistics Plan Summary

Transport, Propellant, Test & Launch Site Operations



HOT FIRE TESTING

Site: Weibert's Auto Place
720 Creek Ln, Youngstown NY

Transport: Team vehicles +
rental truck for test stand

Test campaign: hydrostatic →
cold flow → hot fire

Safety perimeter: 500 ft min
during load & fire

LAUNCH OPERATIONS

Site: FAR — Friends of
Amateur Rocketry
Mojave Desert, CA • ~2,500 mi
from RPI

Transport: Rental box truck;
4-day drive each way

Launch window: April–May
2027

Compliance: Meets all
regulations

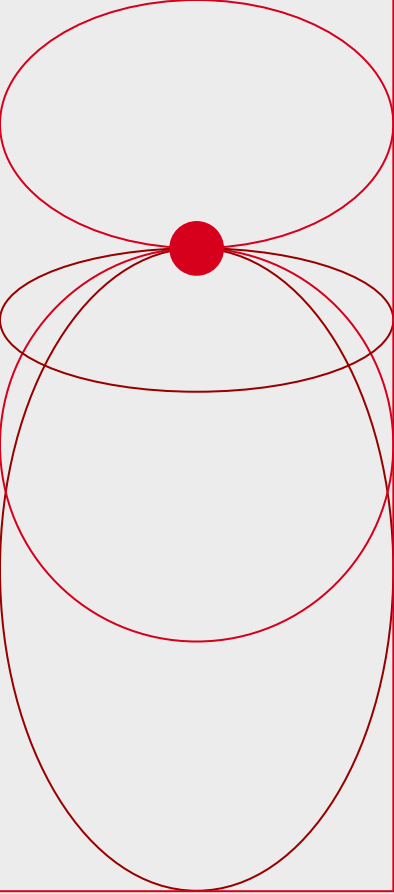
PROPELLANT & LEAD TIME

N₂O: Airgas delivery; ~\$14.4/lb
Ethanol: Supplier; ~\$5–10/lb

FAR-day propellants:
procured on-site, not
transported

Long-lead items: DMLS print
slots, pressure vessels, valves
(8–12 wk lead)

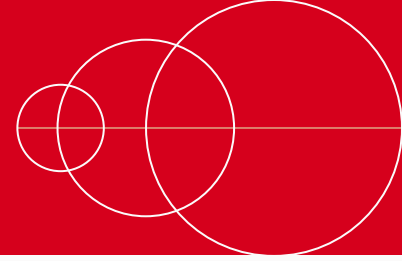
LOGISTICS RISKS TO WATCH: Cross-country transport timing, DMLS print slot availability, FAR launch-slot coordination



Technical Overview

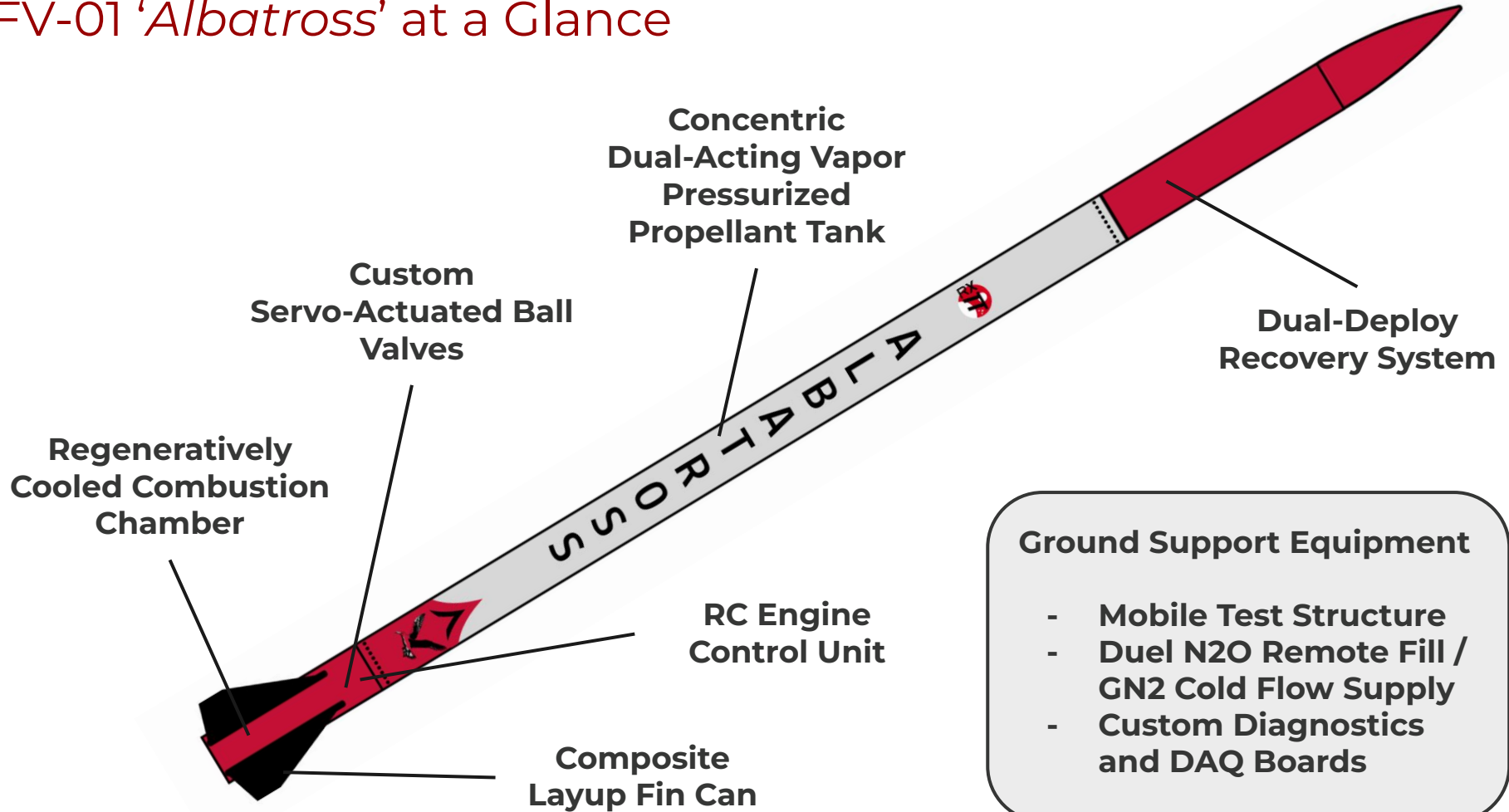
Chief Engineer: Maximilian Slavik

Principal Sources

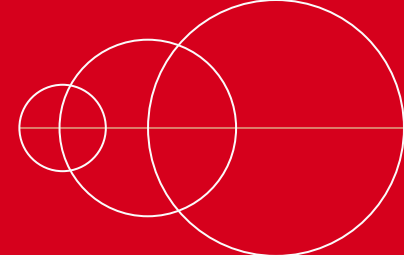


- **Heister, S. D., Anderson, W. E., Pourpoint, T. L., & Humble, R. W.** *Rocket Propulsion*. Cambridge University Press, 2019.
- **Sutton, G. P., & Biblarz, O.** *Rocket Propulsion Elements*, 9th ed. Wiley, 2017.
- **Huzel, D. K., & Huang, D. H.** *Modern Engineering for Design of Liquid-Propellant Rocket Engines*, 2nd ed. AIAA, 1992.
- **Shigley, J. E., Mischke, C. R., & Budynas, R. G.** *Shigley's Mechanical Engineering Design*, 10th ed. McGraw-Hill, 2014.
- **Sharp, C., & Sennott, A.** *Mojave Sphinx Build, Integration, and Launch Guidebook*, HCR-5100, Rev. 01. Half Cat Rocketry, 2024.
- **Parker Hannifin.** *Parker O-Ring Handbook*, ORD 5700. Lexington, KY: Parker Hannifin Corporation.

FV-01 'Albatross' at a Glance



Comprehensive System Modeling



High Burn Time and Active Cooling Necessitate a High Fidelity Model

Propellant Tanks + Feed

- Fill/Drain Transients
- DAVP thermodynamics (piston work and flow work)
- Multiphase Flow

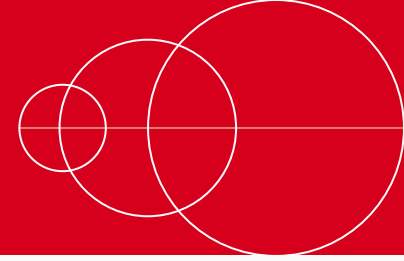
Combustion Phenomena

- Mixture Ratio
- Boundary Layer Effects
- Film Cooling
- Chamber Pressure
- Over/Under Expansion
- Stiffness Variance

Regenerative Cooling

- Darcy Weisbach and Geometry-based dP
- Channel Convection
- Hotwall Thermal Stresses

Comprehensive System Modeling



Propellant Tanks + Feed

- Piston PdV work and flow work, BrentQ root finder for vapor mass fraction and saturation temperature
- Multiphase flow pressure drop / boiling effects

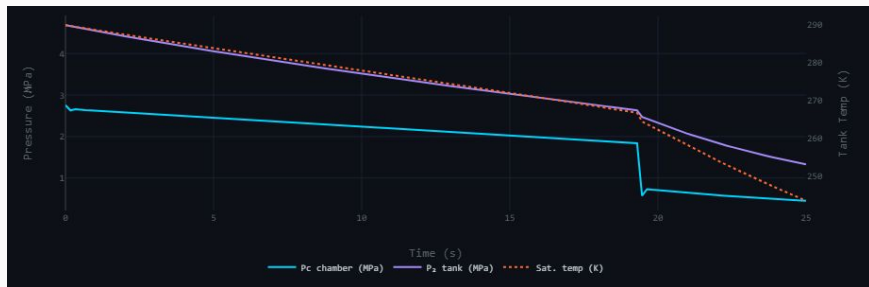
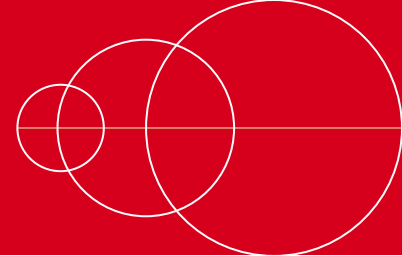
Regenerative Cooling

- Calculation of Re, Pr, HTC's (Gnielinski and Dittus-Boelter), coolant velocities
- Darcy Weisbach pressure drop + geometric change in pressure
- Hotwall temperatures and thermal stresses

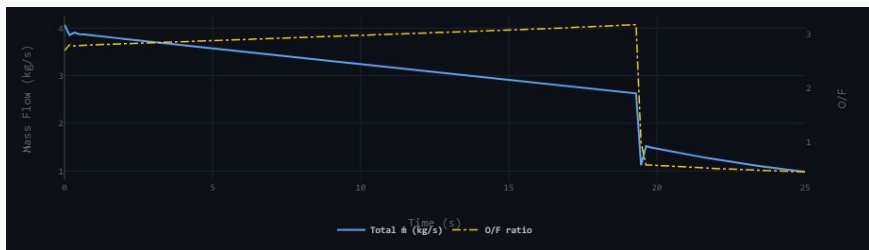
Combustion (Fortran CEA)

- Adiabatic flame temp $\rightarrow$ chamber pressure, C^* , thrust, Isp, stagnation properties
- Chamber transport of nondimensional parameters and stagnation properties
- Boundary layer recovery temperatures and Bartz HT Correlation

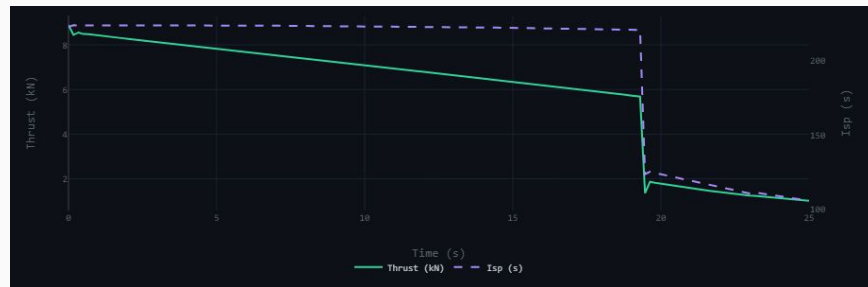
Comprehensive System Modeling



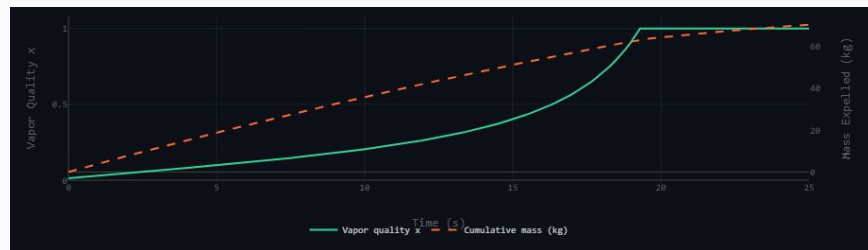
Tank Sat. Pressure and Temperature, Chamber Pressure



Mass Flow Rate and O/F Ratio

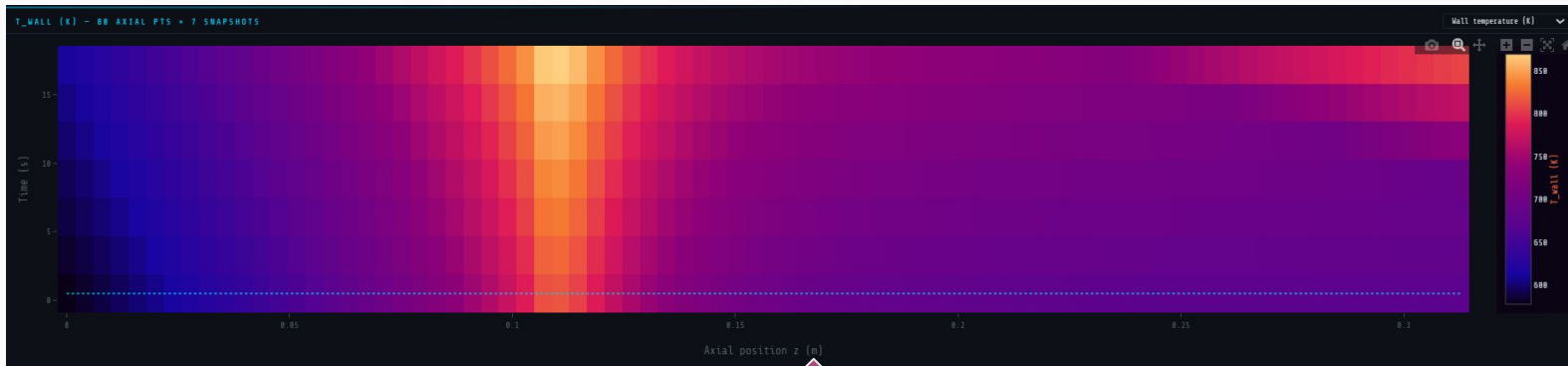
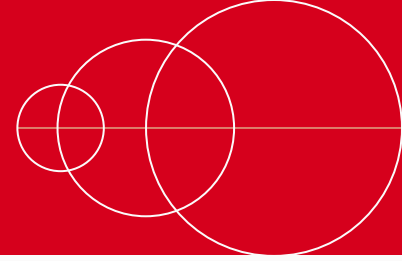


Thrust and Isp (Theoretical Delivered Performance)



Tank Vapor Mass Fraction and Cumulative Expended Mass

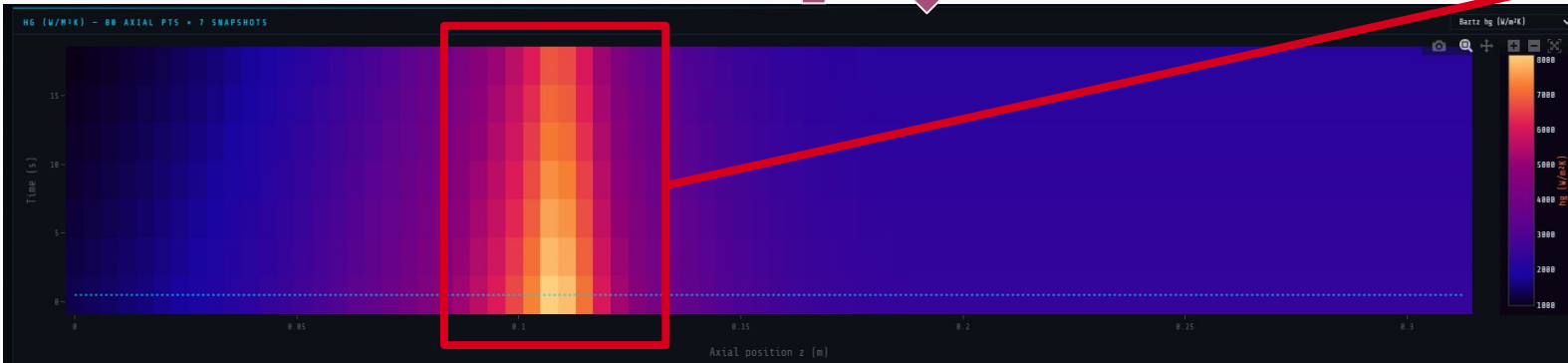
Comprehensive System Modeling



Hotwall Temperature Map (Z-coordinate and Time)

And Bartz HTC (Z-coordinate and Time)

Sharp HTC Peak @ Throat

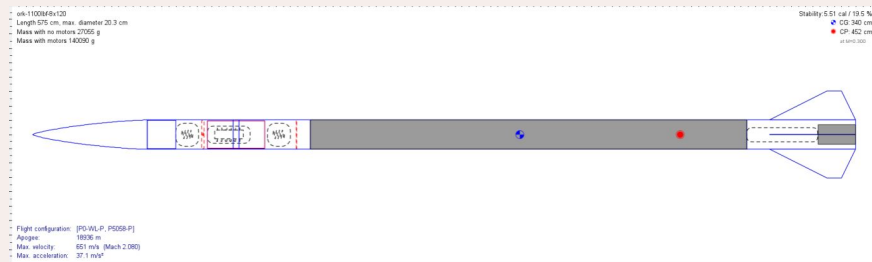


Parameter Selection

Size (Tank Diameter, Length)

Chosen Values: 8" Diameter, 10ft Length

Python *CoolProp* Library → Openrocket sim

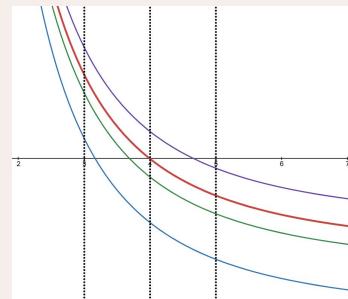


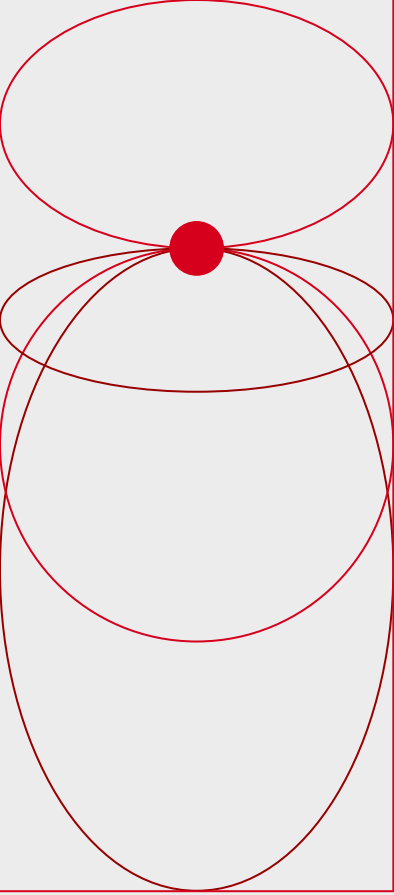
O/F Ratio, T_ambient,
Inner Tank Dimensions

Chosen Values: 3.2 O/F, 63.2F (TARGETS),
4" Internal Tank Diameter

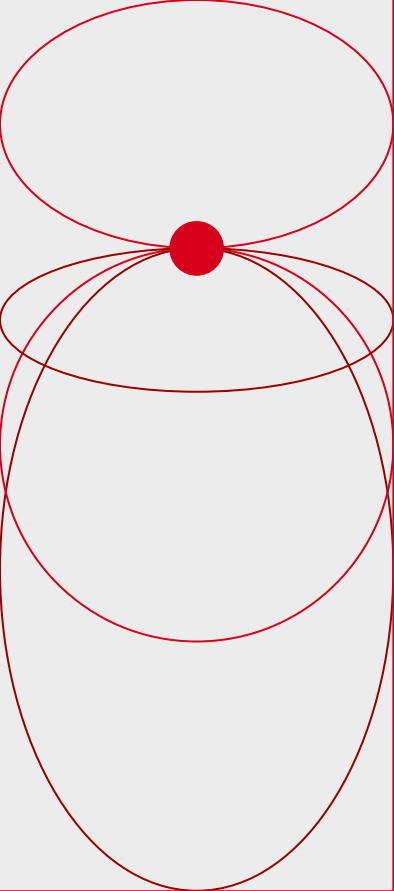
Python *CoolProp* Library → Tank Dimensions
Root Finder + O/F Input Sweep

$$\frac{\left(R_o^2 - \left(\frac{x}{2}\right)^2\right)}{\left(\frac{x}{2} - t\right)^2} - \phi\left(\frac{\rho_f}{\rho_{ox}}\right)$$





Subsystem Engineering Review



Propulsion

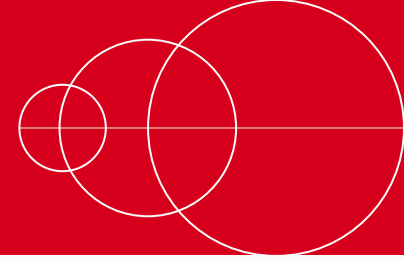
Subsystem Engineering Review

Subsystem Lead: Maximilian Slavik + Reese Mount

Subsystem Members: Yeshe Dorje Yodsampa, Eli Macdonald

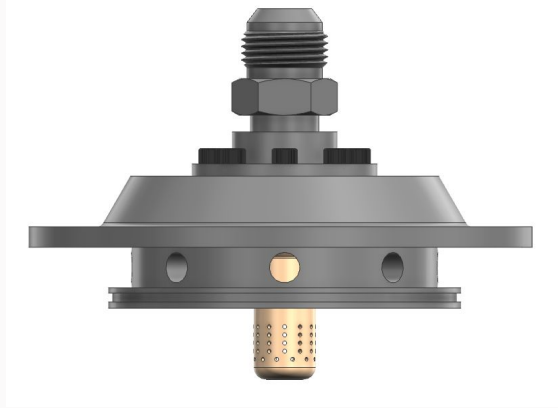
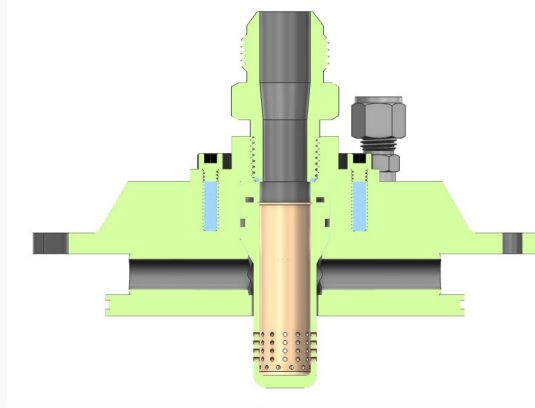
Pintle Injector

Responsible Engineer: Yeshe Dorje Yodsampa



Component Criteria:

- $dP/P_c > 20\%$
- 3.2 O/F Ratio
- Account for NHNE Effects
- Vertical Slots - Shear Instability Effects
- Elastomeric Seals
- Centering Feature (s)
- 13-15% fuel film cooling

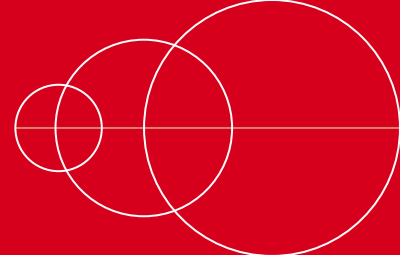


Further Development

Usage of NHNE Modeling code to analyze geometry improvements, detailed film cooling effectiveness models, ignition thermal transience

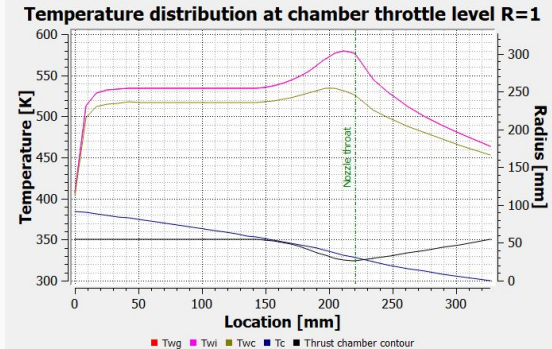
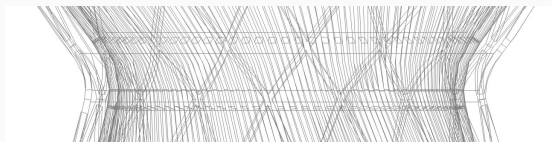
DMLS Combustion Chamber

Responsible Engineer: Maximilian Slavik



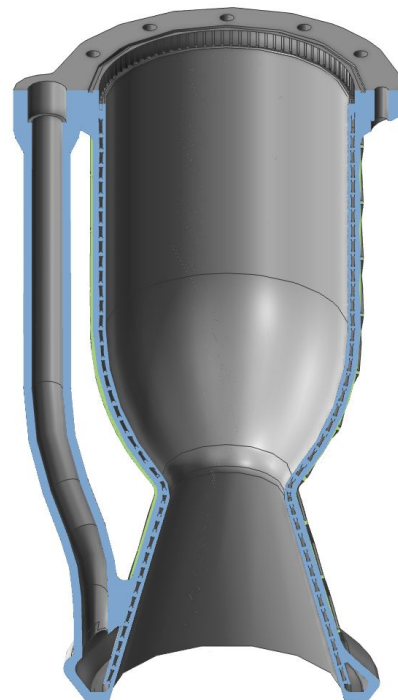
Component Criteria:

- Al10SiMg and DFAM
- Maximum Hotwall Temp 600 K
- Mitigation of Volute Static Pressure Variations
- Chamber $L^* > 0.85\text{m}$
- Nozzle Expansion to 0.7Bar



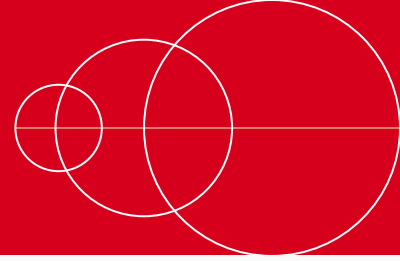
Further Development

Further channel and volute shape optimization, examine multi-piece L^* extensions, refine thermal stress modeling



Cold Flow Manifold

Responsible Engineer: Yeshe Dorje Yodsampa



Component Criteria:

- Emulate regenerative cooling outlet flow
- Avoid restriction of injector resultant jet visibility
- Elastomeric seals

Further Development

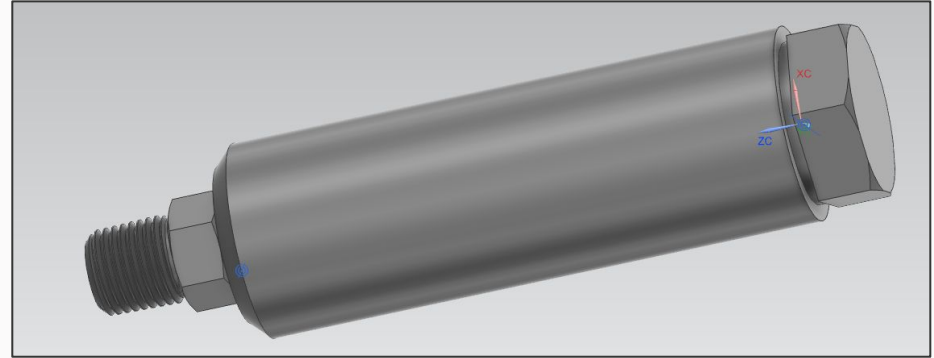
Enhance machinability, CFD modeling of static pressure variations

Head-End Cartridge Igniter

Responsible Engineer: Elias Macdonald, Maximilian Slavik

Component Criteria:

- Provide hot gas manifolding for C-level rocket motor exhaust
- Safely contain hot gas during and after ignition
- Reliable electronic match clearance pathway
- Machinable in-house



Further Development

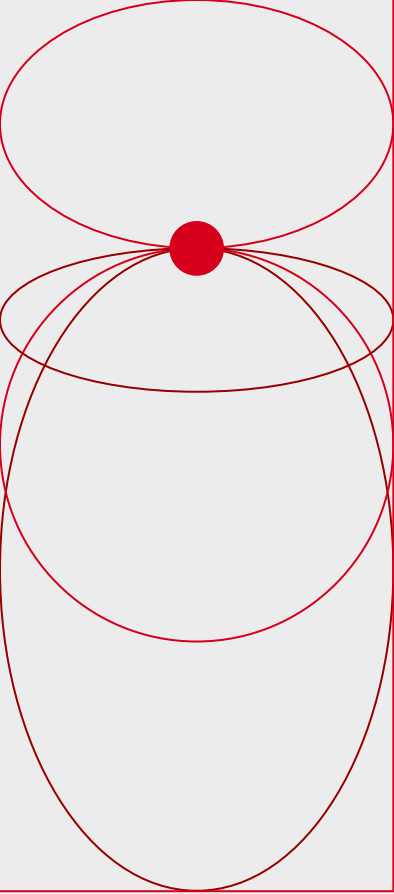
Transient heat transfer modeling, replaceability optimization



Notable Interfaces - Propulsion

Relevant Components	Importance (1→3)	Brief Explanation
Fuel Inlet → Lower Fluids	3	Fuel Main Line Connection
N2O Inlet → Lower Fluids	3	Ox Main Line Connection
PTs → DAQ Board	1	Static Fire Pressure Data Acquisition
TCs → DAQ Board	1	Static Fire Regen Data Acquisition
Injector → Longeron Brackets	3	Flight Thrust Transfer
Igniter → Ematch+GSE	2	Ignition Feedthrough (Injector)





Fluids

Subsystem Engineering Review

Subsystem Lead: Slater Selinger

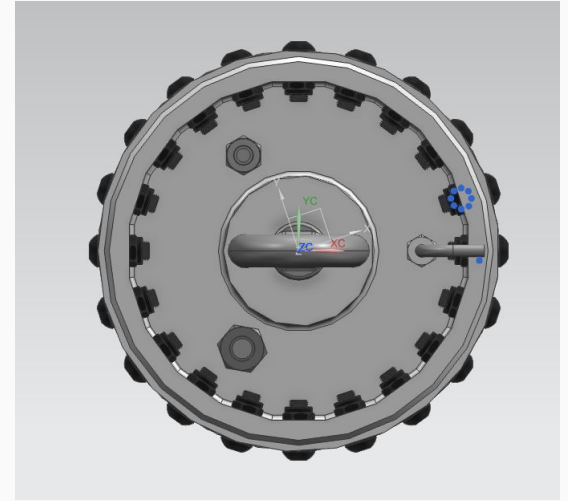
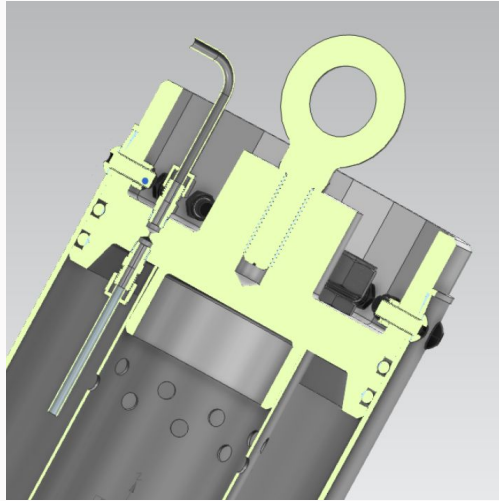
Subsystem Members: Ty Hollis, Ryan Ing, James Smith, Ben Wahl,
Keelan Wheeler

Upper Bulkhead

Responsible Engineer: Slater Selinger, Ty Hollis

Component Criteria:

- Allow for Ethanol Pressurization
- Hold pressurized N₂O
- Slow Venting into atmosphere

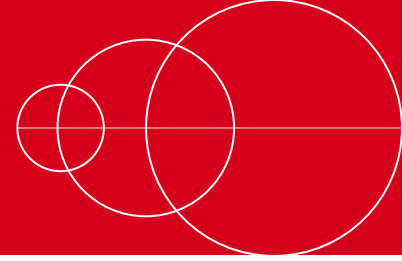


Further Development

Further information into the proper hole sizing from other teams, and a full numerical analysis of the eyebolt

Concentric Tanks

Responsible Engineer: Ty Hollis

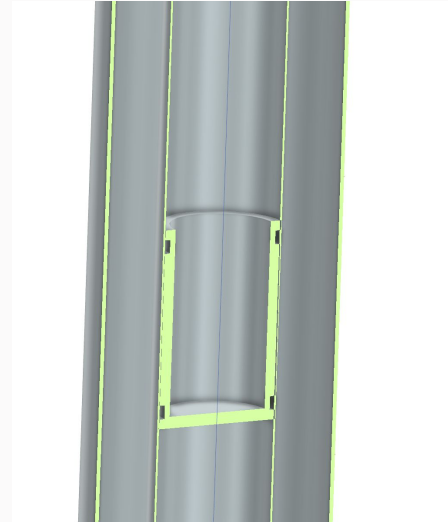


Component Criteria:

- Hold pressurized N₂O and Ethanol
- No liquid leakage
- Interfacing with both bulkheads

Further Development

Perform thermal and structural FEA analysis

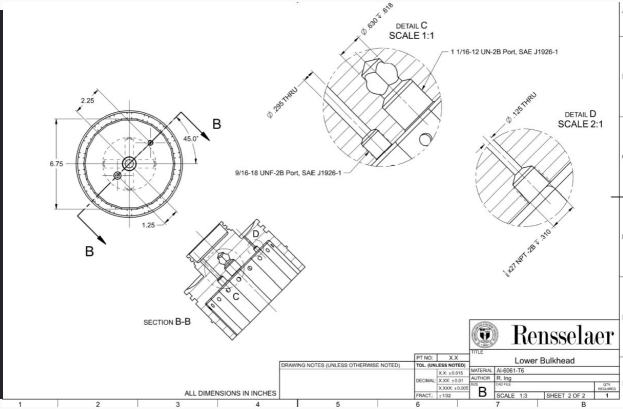
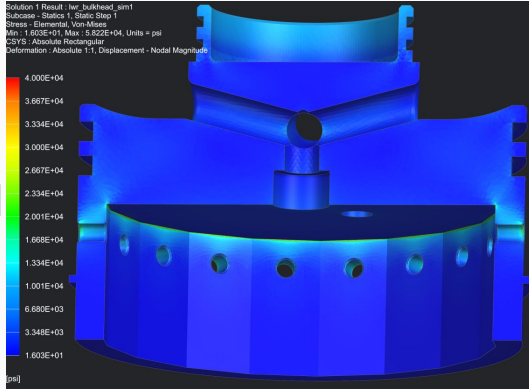


Lower Bulkhead

Responsible Engineer: Ryan Ing

Component Criteria:

- Input and output for N₂O and Ethanol
- Interfacing with Tanks and Aeroframe
- Inner seal for no leaks



Further Development

Talking with Sam Chiappone about manufacturing and perform thermal stress modeling

Servo Actuated Ball Valve (SABV) and Feed Lines

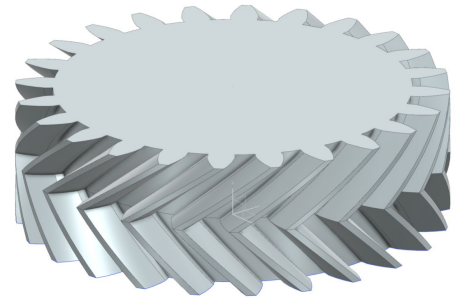
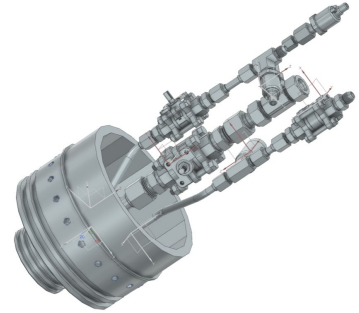
Responsible Engineer: *Keelan Wheeler*

Component Criteria:

- Movement of all fluids from lower bulkhead to manifold
- Proper McMaster Carr parts and correct modeling of pipes/valves
- SABVs have minimal force to open
- SABVs fully control flow of fluids

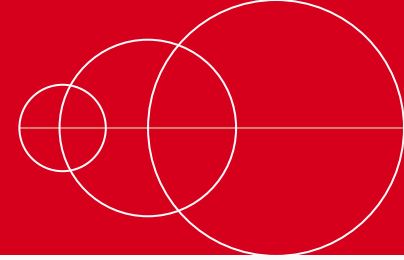
Further Development

Structural analysis of gear and ensuring SABV's have full operational capacity



Pressure Piston

Responsible Engineer: Ben Wahl

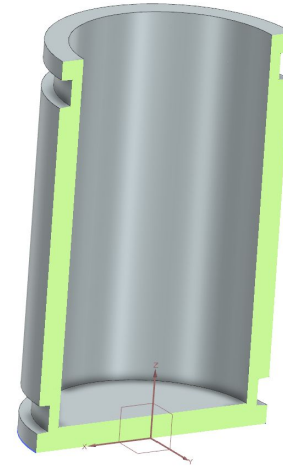


Component Criteria:

- Transference of pressure from N₂O to ethanol
- Cannot leak at all, most important seal on the entire body
- Does not rack to prevent pressure transfer

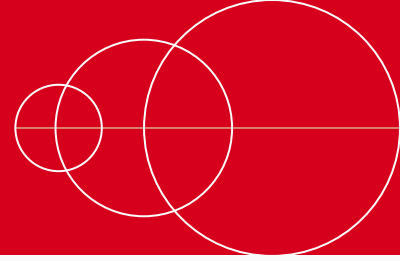
Further Development

Numerical simulation to prove the lack of racking and the proper transference of pressure to the ethanol



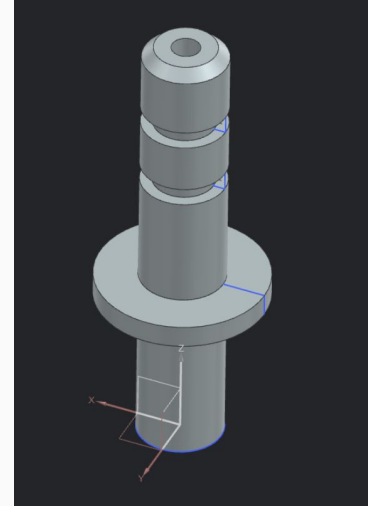
Quick Disconnect Valve

Responsible Engineer: James Smith



Component Criteria:

- Instant disconnect for N2O Inlet
- No leakage of N2O
- Interfacing with GSE for N2O fill
- Holding at 1000 psi

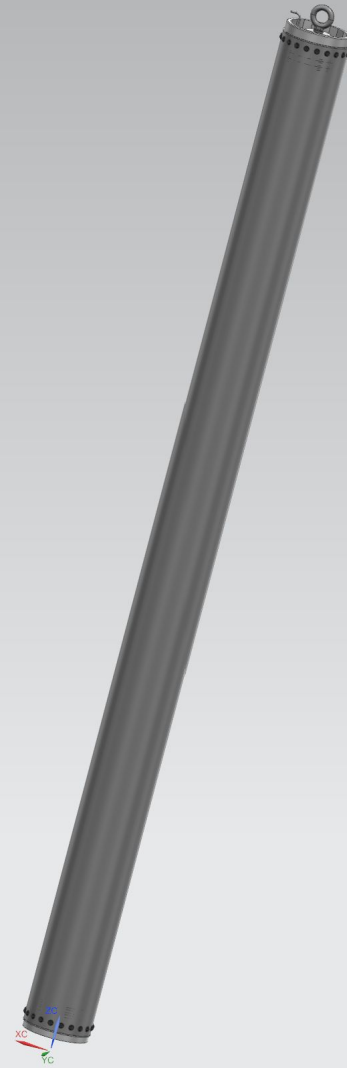


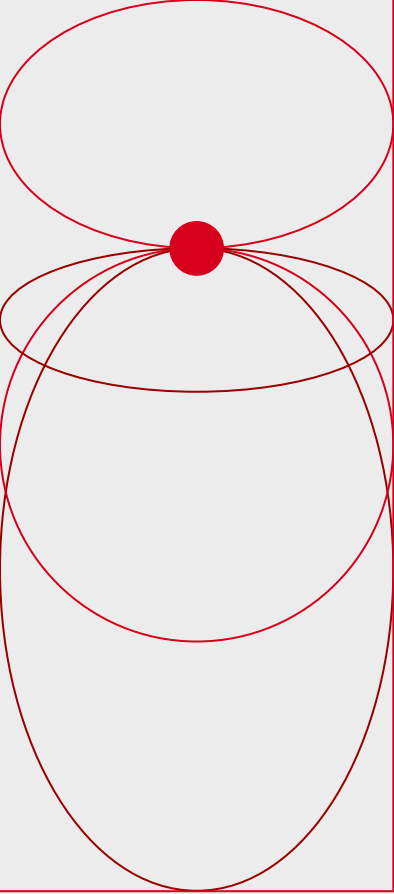
Further Development

Looking into other QD components and comparing them to current model.

Notable Interfaces - Fluids

Relevant Components	Importance (1→3)	Brief Explanation
Lower Bulkhead → GSE fill station	3	Filling of N2 for testing and N2O for hot firing
Lower Bulkhead → Manifold	3	N2O and Ethanol to manifold
Lower Bulkhead → Aeroframe	3	Structural housing between manifold and lower bulkhead
Upper Bulkhead → GSE Fill	3	N2 Fill for cold flow
Upper Bulkhead → Aeroframe	3	Structural Connection of Bulkhead to Aeroframe
Sensors: TC and PT → Systems	2	Data and power connection for static firing testing
SABVs→ Electrical Power	2	Open/Close for fuel flow





Airframe & Recovery

Subsystem Engineering Review

Subsystem Lead: Riley Dunn

Subsystem Members: Eli Corr, Theo Kraemer, Daniel Xia, Matt Sweitzer, Matt Fuller, Zen Pinkenburg

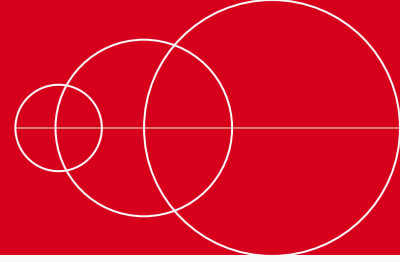
Component Criteria:

- ## Further Development

- | <i>Parachute Trade Study (Pugh Matrix)</i> | | | | | | |
|--|----------------|----------|-------------------|----------|----------|----------|
| Project: RPU-2 | | | | | | |
| Document ID: | | | | | | |
| System: Recovery | | | | | | |
| Date: 4/12/2026 | | | | | | |
| Status: Completed | | | | | | |
| Author/Responsible Engineer: Eli Corr | | | | | | |
| Decision: 30 ft Standard Parachute | | | | | | |
| Scoring Matrix | | | Option Definition | | | |
| Weight: Assign 1-5 (5 = Critical, 1 = Nice to Have) | | | Baseline: | | | |
| Score: Compare to Baseline. (+) Better, (-) Same, (-) Worse. | | | Option 1: | | | |
| A raw 1-5 score scale is also an alternative. | | | Option 2: | | | |
| <i>Weights & Values represented in the template are an example.</i> | | | Option 3: | | | |
| <i>The Analysis & Rec. can be copy & pasted to the Rec. of the DDR</i> | | | Option 4: | | | |
| Evaluation Criteria | Weight (1 - 5) | Baseline | Option 1 | Option 2 | Option 3 | Option 4 |
| Volume | 2 | o | 1 | -1 | -1 | 1 |
| Landing Speed | 4 | o | -1 | 1 | 1 | -1 |
| Mass | 1 | o | 1 | -1 | -1 | 1 |
| Cost (\$) | 5 | o | 1 | -1 | -1 | -1 |
| Line Strength | 4 | o | -1 | 1 | 1 | 1 |
| | o | o | o | o | o | o |
| | o | o | o | o | o | o |
| TOTAL SCORE (+) | | o | 3 | 2 | 2 | 3 |
| TOTAL SCORE (-) | | o | 2 | 3 | 3 | 2 |
| NET WEIGHTED SCORE | | o | 1 | 0 | 0 | -2 |
| RANK | | 1 | 1 | 1 | 1 | 5 |

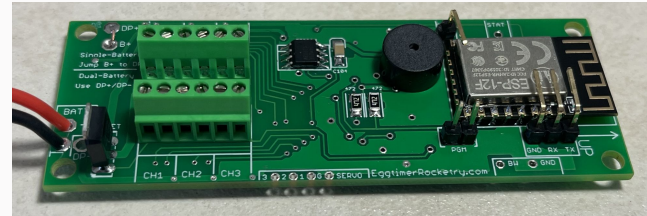
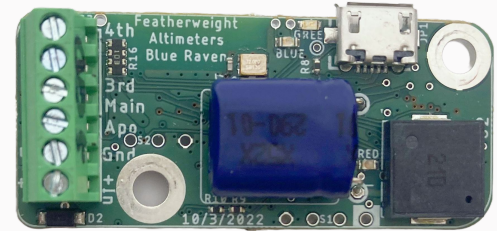
Recovery Avionics

Responsible Engineer: Eli Corr



Component Criteria:

- “Blue Raven” primary altimeter
- “EggTimer Proton” secondary altimeter
- “Swift” GPS tracker w/ self-contained ground station
- Independent batteries for all 3
- Physical arming switches
- Redundant BP sep. charges

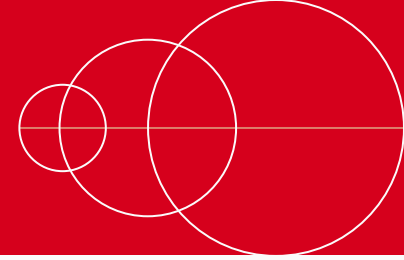


Further Development

- BP charge sizing & consideration of high-altitude effects
- Pressure sampling port count & sizing

Upper Airframe

Responsible Engineer: Daniel Xia



Component Criteria:

- 5:1 Tangent Ogive G-12 nose cone w/ 8" coupler
- 2x 30" body tubes (G-12 fiberglass), exact length pending volume reqs.
- 18" G-12 Avionics Bay
- Screw circle from upper body tube to avbay, & middle tube to upper tank bulkhead
- Shear pins from nose cone to upper tube, & avbay to middle body tube

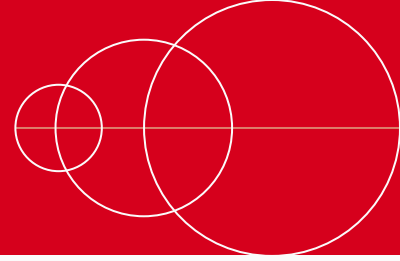
Further Development

- Shear pin count, sizing, & stress analysis
- Exact volume estimation for BP charges
- Pressure vent port count & sizing



Fin Can Assembly

Responsible Engineer: Matt Sweitzer

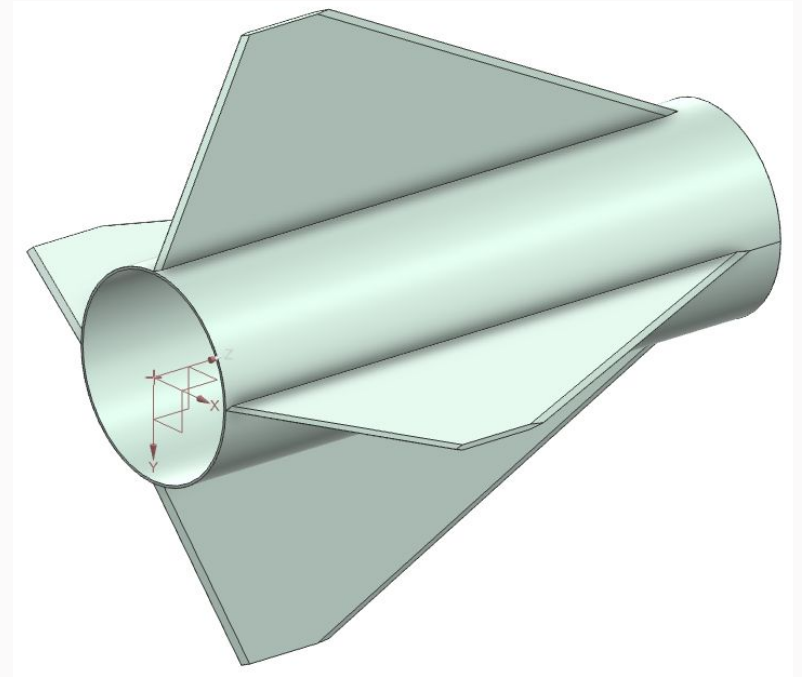


Component Criteria:

- 4x 5mm G-12 fiberglass fins, carbon fiber tip-to-tip layup
- Static stability margin:
 $(X_{CP} - X_{CG})/D = 4.98 \rightarrow 3.47$
- Fin flutter speed always remains $\geq 1.5x$ velocity
- Clipped Delta profile w/
Hexagonal cross-section

Further Development

- Finalize dynamic stability investigation
- Then run drag optimization given stability constraints



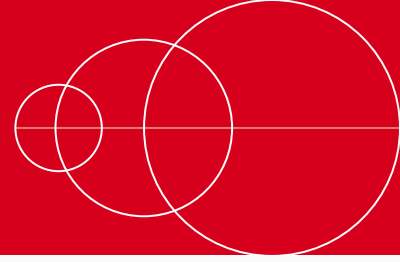
Fin Trade Studies

Fin Planform Profile Trade Study (Pugh Matrix)							Str
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Fin Cross-Section Trade Study (Pugh Matrix)							
Project:		Aquila					
Document ID:							
System:		FV-01 Albatross - Airframe & Recovery					
Date:		3/24/2026					
Status:							
Author/Responsible Engineer:		Matt Switzer					
Decision: Hexagonal							
Scoring Matrix				Option Definition			
Weight: Assign 1-5 (5 = Critical, 1 = Nice to Have)				Baseline: DoubleWedge			
Score: Compare to Baseline. (+) Better, (o) Same, (-) Worse.				Option 1: Hexagonal			
A raw 1-5 score scale is also an alternative.				Option 2: BluntTrailingEdge			
Weights & Values represented in the template are an example.				Option 3: Tapered			
The Analysis & Rec. can be copy & pasted to the Rec. of the DDR				Option 4: Rounded			
Evaluation Criteria	Weight (1 - 5)	DoubleWedge	Hexagonal	BluntTrailingEdge	Biconvex(Airfoil)	Rounded	
Performance (Aerodynamics)	5	o	o	-1	-1	-1	
Manufacturability	5	o	o	o	-1	-1	
Structural Durability	4	o	1	1	1	1	
		o					
		o					
		o					
		o					
TOTAL SCORE (+)		o	1	1	1	1	
TOTAL SCORE (-)		o	o	1	2	2	
NET WEIGHTED SCORE		o	4	-1	-6	-6	
RANK	2		1	3	4	4	
Trade Study Result: Hexagonal							
Analysis & Recommendation							
The Diamond (double wedge) and Hexagonal designs were found to be the most commonly used for supersonic rockets, with the hexagonal design possibly being slightly more structurally sound and resisting warping or breaking on impact better in some cases. The blunt trailing edge design has multiple pros and cons to the aerodynamic performance, where it reduces the strength of shockwaves (the main source of drag at faster speeds) but also a base level of drag as the squared off back end creates a wake. It should be noted that the "higher speeds" mentioned were transonic speeds, so I'm unsure if it would have the same efficiency at supersonic speeds. The biconvex option seemed to perform better at slower speeds, and I found that they are generally harder to manufacture as well, since the "airfoil" shape must be symmetric or else there will be issues and greater drag. As far as the rounded shape, it is seen in OpenRocket as an option for fin cross-section, but as far as I can tell, it is sort of the same as the biconvex design, so I treated it the same.							

Thrust Transmission Assembly

Responsible Engineer: Theo Kraemer



Component Criteria:

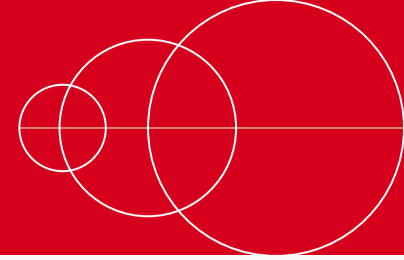
- Transmit 1500 lbf thrust from engine to tank bulkhead
- Resists buckling w/ factor of safety of 1.5 at 48" maximum length
- Provide support for fluids & electronics components
- C-channel 0.05" sheet metal with 1.75 K-factor

Further Development

- Select COTS brackets & fasteners
- Design support "rack" for electronics components

Centering Ring

Responsible Engineer: Zen Pinkenburg

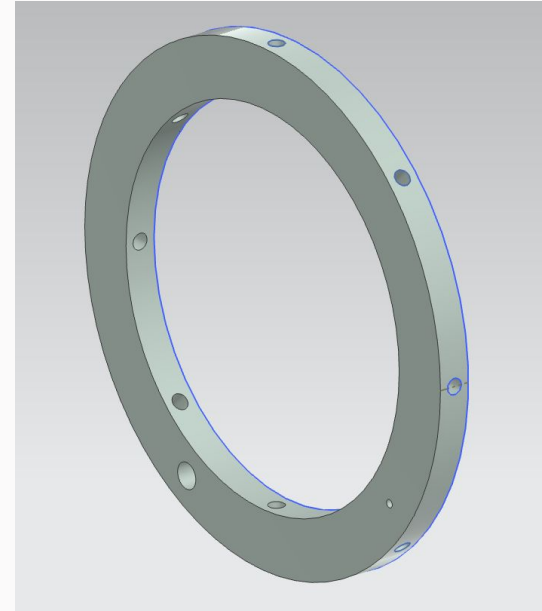


Component Criteria:

- Interface between Thrust Transmission Assembly, fin can, and Fluids/ES QD
- Transmit aerodynamic force from fins to primary airframe structure

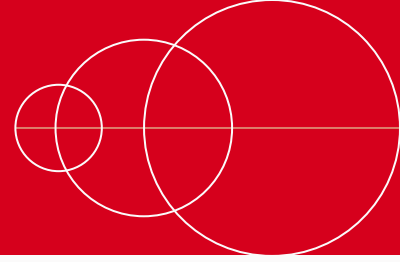
Further Development

- Finalize QD hole sizing & placement
- Identify COTS brackets & fasteners for TTA and fin can
- Verify stress under expected aerodynamic loading



Launch Rail Guides

Responsible Engineer: Riley Dunn

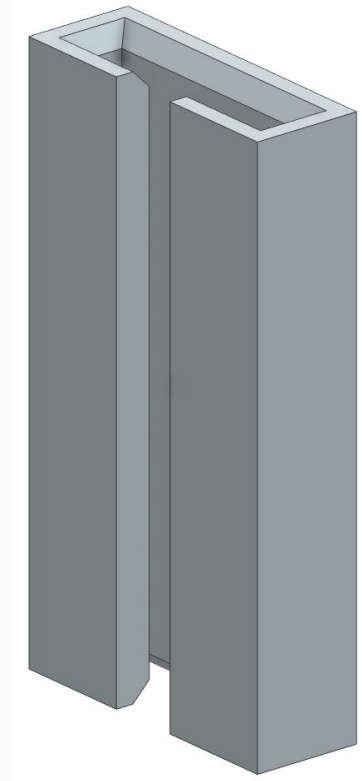


Component Criteria:

- Constraint motion while accelerating on launch rail
- One located ahead of CoM (upper tank bulkhead), one behind (lower BH)

Further Development

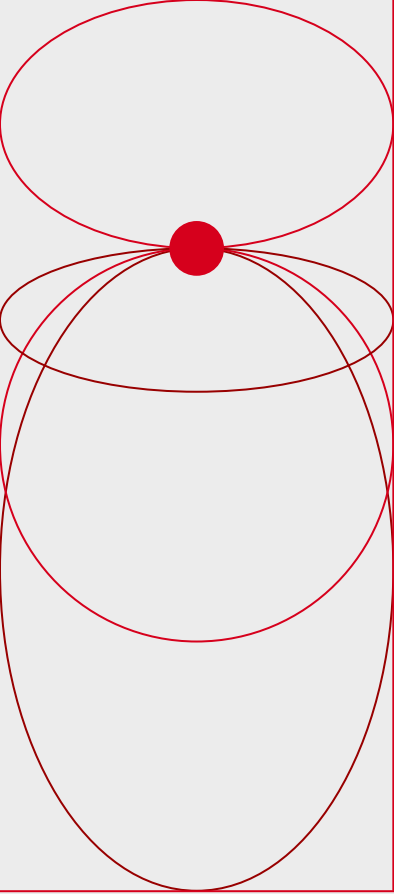
- Load & stress calculation → bolt sizing
- Aerodynamic shaping (Haack-esque profile)



Notable Interfaces - A&R

Relevant Components	Importance (1→3)	Brief Explanation
Thrust Transmission Structure → Lower Tank Bulkhead	3	Longerons bolted to bulkhead, must support full thrust load
Thrust Transmission Structure → Engine Manifold	3	Brackets to longerons, must support full thrust load
Launch Rail Guides → Upper & Lower Tank Bulkheads	3	Bolt onto bulkhead “tabs” bulkheads, slots in body tubes; very high possible load
Recovery Harness → Upper Tank Bulkhead	2	Eyebolt threaded into bulkhead, Nomex wrap around vent tube
Body Tube → Tank Bulkheads	2	Bolt circle (or screws) to bulkhead
Centering Ring → QD fluid & electrical connections	1	Holes through ring
Thrust Transmission Structure → Engine Electronics Assembly	1	“Rack” to support electronics, bolts to longerons





Ground Support Equipment

(Static Fire Test Stand)

Subsystem Engineering Review

Subsystem Lead: Jasper Heymann

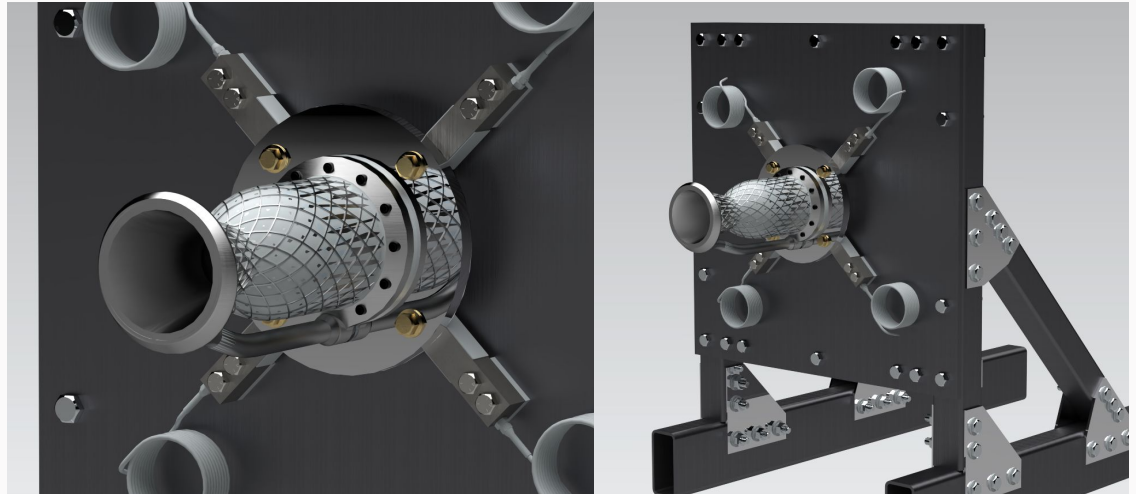
Subsystem Members: Joseph Sardano, Josh De Cesare, Jack Neeleman

Thrust Stand

Responsible Engineer: Jasper Heymann, Josh De Cesare

Component Criteria:

- Securely mount engine horizontally
- Withstand 4000 lbs of thrust
- 4 shear-beam load cells
- Bolted joints only for easier on-site assembly
- Designed with future engines in mind

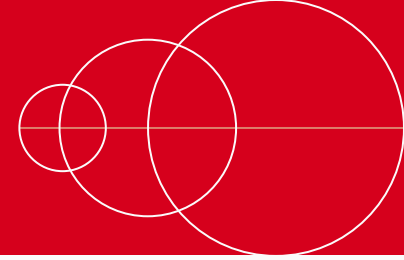


Further Development

FEA tests on individual truss elements (to qualify tube thickness,
FEA tests on overall structure to qualify min FOS = 2.0

Fluids Tower

Responsible Engineer: Joseph Sardano, Josh De Cesare



Component Criteria:

- Secures Fluids tank to rectangular beam
- Measures tank weight with 2 S-beam load cells
- Rotates between horizontal/vertical orientations
- Has two routing paths for N₂O and N₂ loading from remote fill panel

Further Development

Find/add adequate quick disconnects, incorporate final version of fluids tank with appropriate valves/fittings/etc.

FEA to determine max deflection and qualify min FOS = 2.0.



Remote Fill Panel

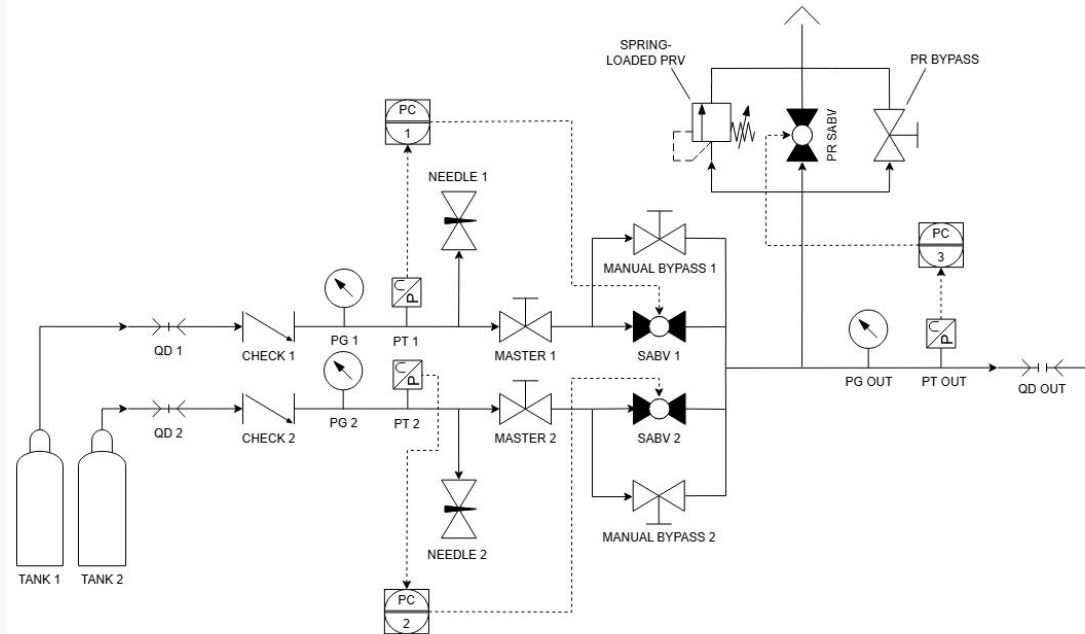
Responsible Engineer: Jasper Heymann

Component Criteria:

- Fill tank via remote and on-panel control
- N20 and N2 at 800 psi continuous
- Quick-disconnect input/output
- FOS 1.5-2.0 for all components
- JIC connections where possible

Further Development

Find/add adequate quick disconnects rated for 800+ psi, add mounting details to model, finalize PID, add electronics box to panel.



Remote Fill Panel

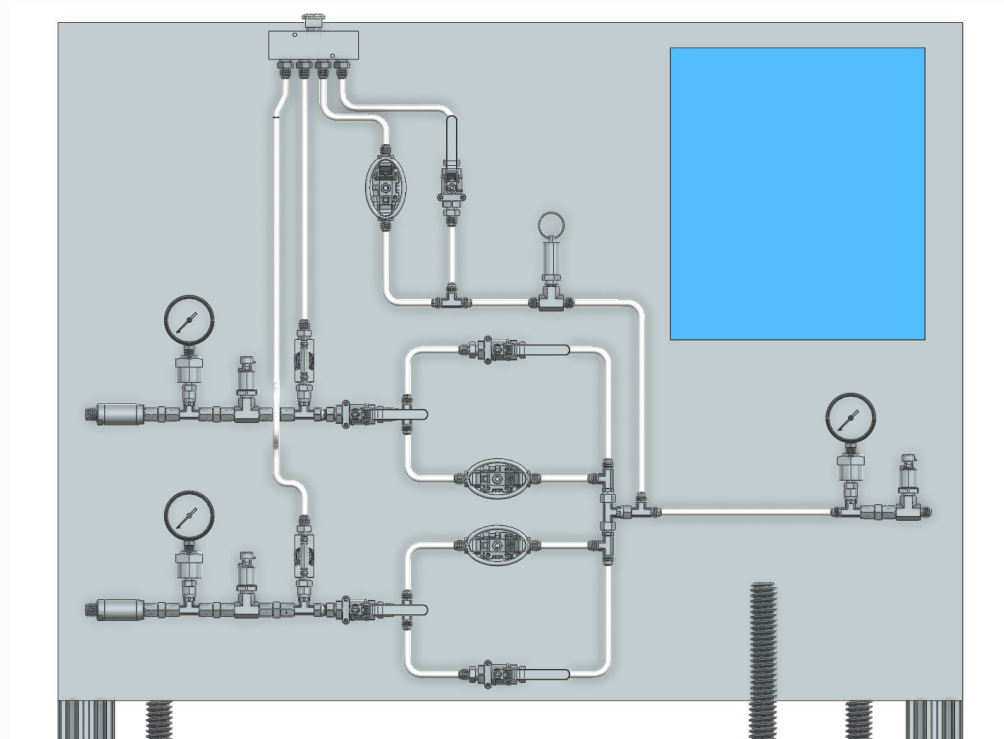
Responsible Engineer: Jasper Heymann

Component Criteria:

- Fill tank via remote and on-panel control
- N2O and N2 at 800 psi continuous
- Quick-disconnect input/output
- FOS 1.5-2.0 for all components
- JIC connections where possible

Further Development

Find/add adequate quick disconnects rated for 800+ psi, add mounting details to model, finalize PID, add electronics box to panel.



Full Assembly

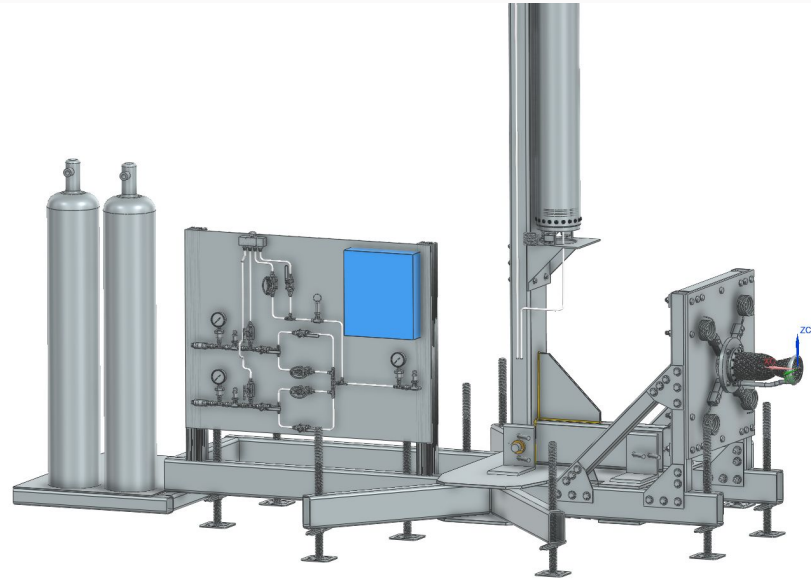
Responsible Engineer: Jasper Heymann, Joseph Sardano, Josh De Cesare, Jack Neeleman

Component Criteria:

- Prioritize ease of assembly on-site
- Combines all subsystems into one master assembly
- Structure secured via tensioned chains attached to ground-anchored stakes

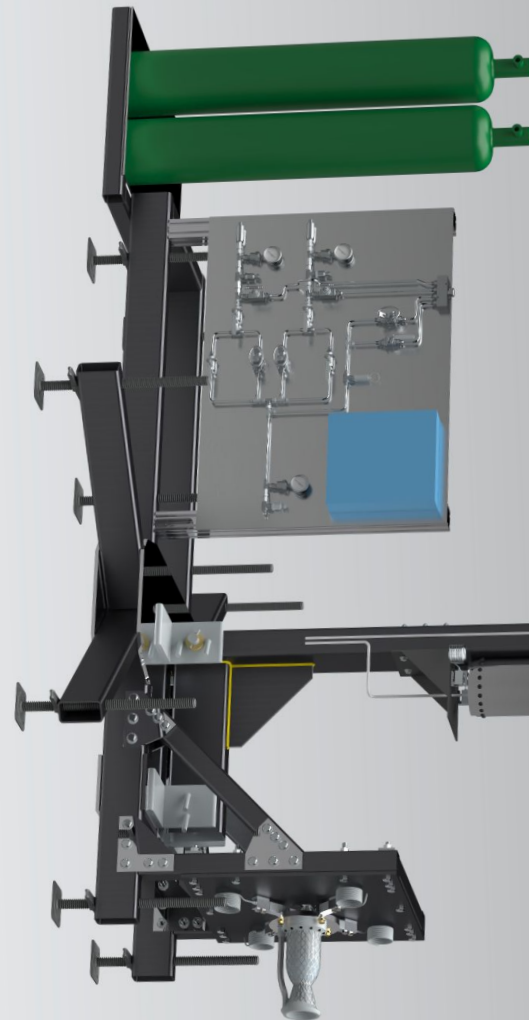
Further Development

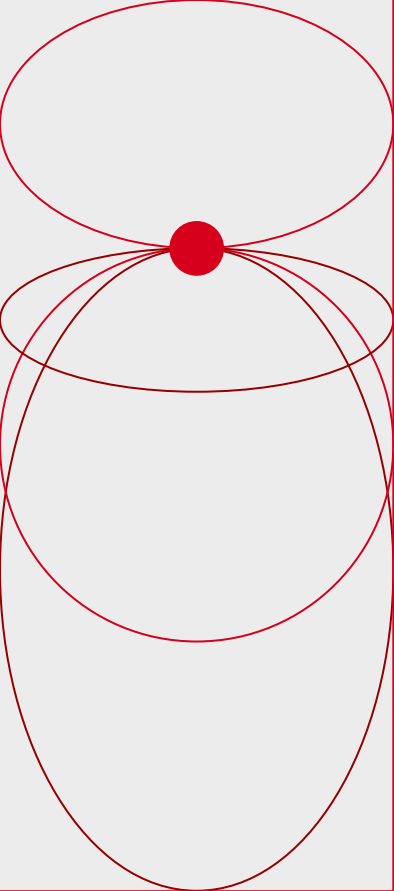
Finalize remaining assembly details (extra bolts, k-bottle holder, etc.). Find adequate quick disconnects and determine/model flexible hose lengths. Incorporate final models of subsystems (prop, electronics, fluids), add pulley system.



Notable Interfaces - GSE

Relevant Components	Importance (1→3)	Brief Explanation
Remote Fill → Lower/Upper Fluids	3	Filling of N2 for testing and N2O for hot firing
SABVs → DAQ/DCS	3	Remote fill SABVs controlled by DCS
Pressure Sensors → DAQ/DCS	3	Remote fill pressure sensors feed data to DAQ/DCS
Engine Load Cells → DAQ/DCS	3	4 shear-beam load cells feed data to DAQ/DCS
Fluids Tank Load Cells → DAQ/DCS	3	2 S-beam load cells feed data to DAQ/DCS
Fluids Tank & Engine → Load Cells	2	Tank - 2 S-beam load cells Engine - 4 shear beam cells





Electronics & Software

Subsystem Engineering Review

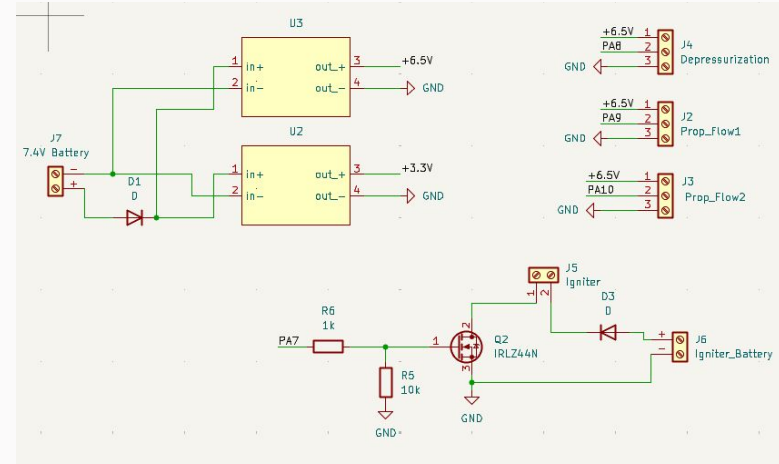
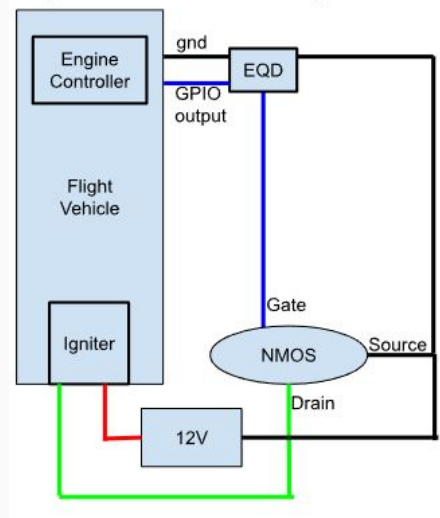
Subsystem Lead: Sydney Howell

Subsystem Members: Cooper Werner, Saanvi Suresh, Aidan Miceli
Nicholas Gladu, Samson Kempiak, Anthony Santiago, Milana
Smirnoff

Engine Controller - Hardware

Criteria:

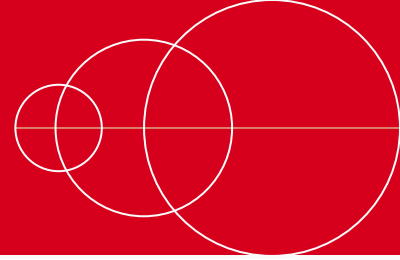
- Control propellant flow
- Ignite the engine
- Depressurize the engine
- Must fit into lower fluids bay
 - 5" x 3" x 1.5"



Further Development

Test current draw from SABVs, finalize power supply circuitry, design a custom PCB, design the EQD

Engine Controller - Firmware

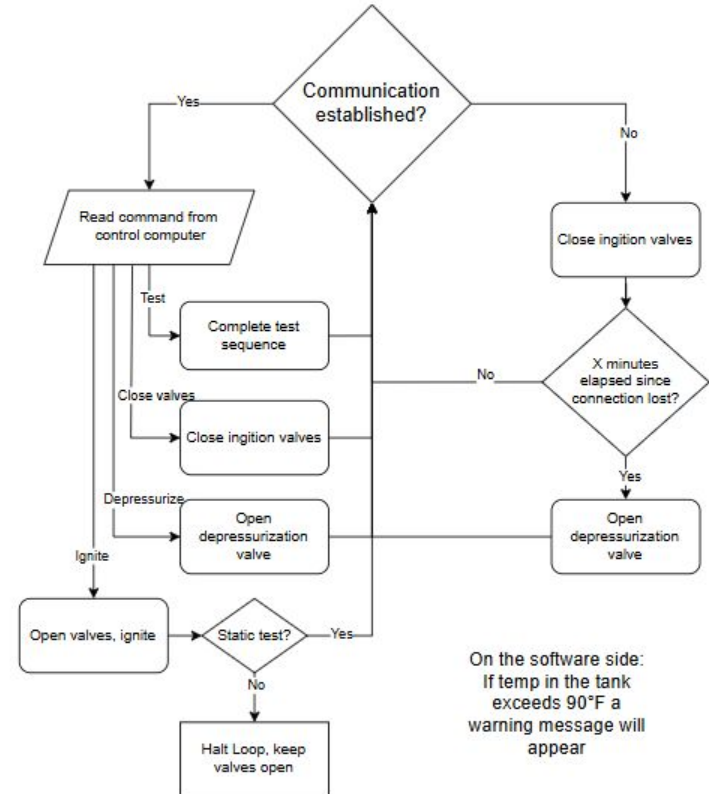


Criteria:

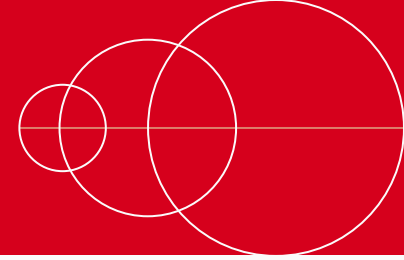
- Depressurize if communication is lost
- Take the following commands:
 - Test Sequence
 - Ignite
 - Close valves
 - Depressurize

Further Development

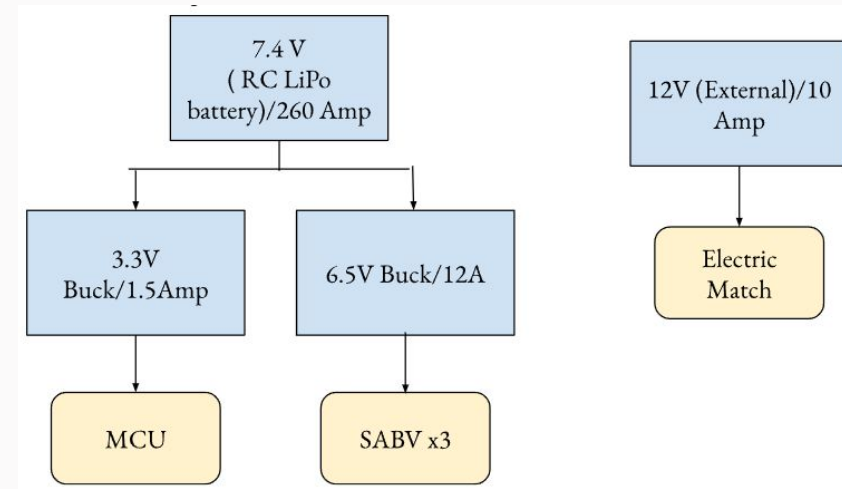
Finalize command requirements, test PWM generation



Engine Controller - Power



Component	Voltage	Current	Quantity
Microcontroller	3.3 V	<300 mA	1
Servo (40kg)	4.8V-6.8 V	<3.9A	2
Servo (80kg)	6V-8.4V	<6.5A	1
*Electric match	>3V	1A	1



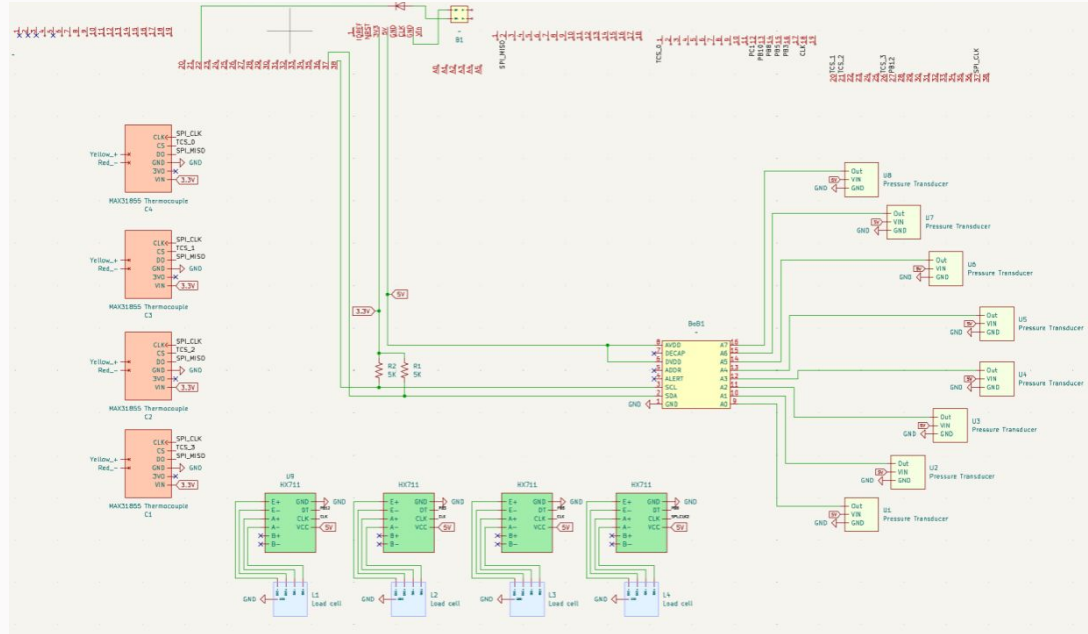
Data Acquisition (DAQ) - Hardware

Criteria:

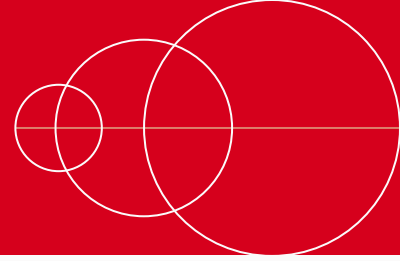
- Collect data during static tests
- 2 thermocouples measure engine temperature
- 4 load cells measure thrust
- 3 pressure transducers in the combustion chamber
- 1 pressure transducer in the tank

Further Development

Test ADC breakout boards, design a custom PCB



Data Acquisition (DAQ) - Firmware

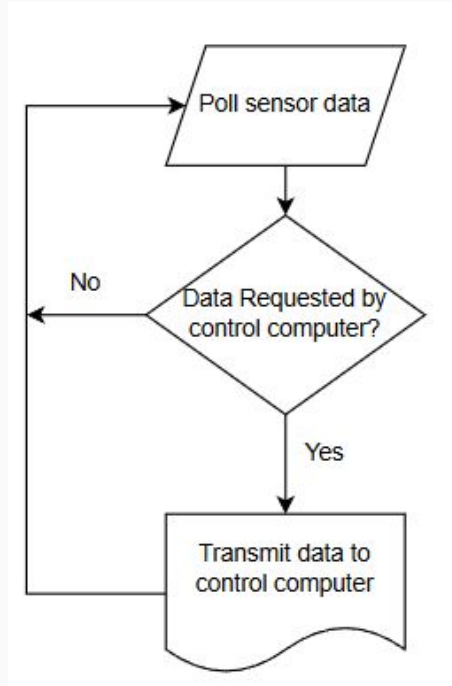


Criteria:

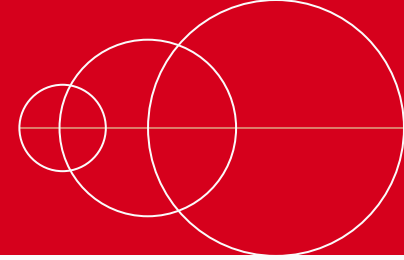
- Collect and transmit data from sensors
 - Temperature (SPI)
 - Pressure (I2C)
 - Thrust (Custom HX711 communication protocol)

Further Development

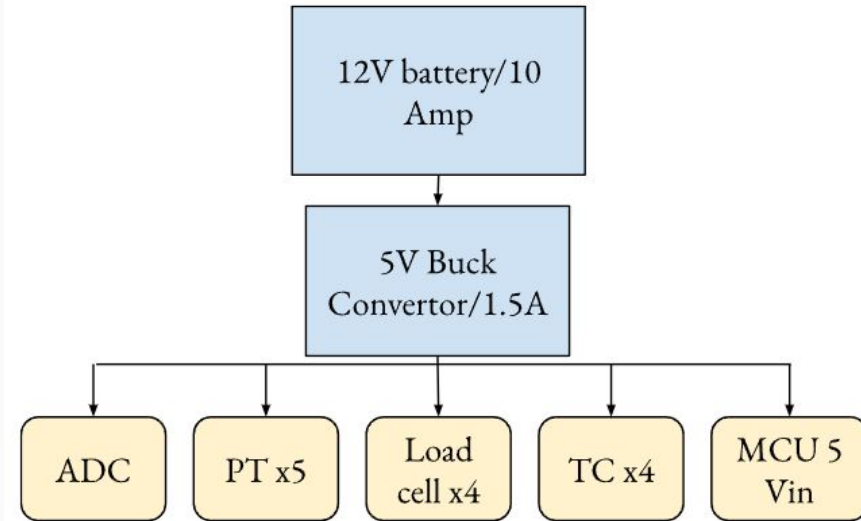
Test microcontroller I2C and SPI functionality, test and calibrate sensors



Data Acquisition (DAQ) - Power



Component	Voltage	Current	Quantity
Microcontroller	3.3 V	<300 mA	1
ADC	5V	0.2mA	1
HX711/Loadcell	5V	10-30mA	4
MAX31855 (TC)	3.3V-5V	1500 μ A	4
Pressure Transducer(PT)	5V	4-20mA	5



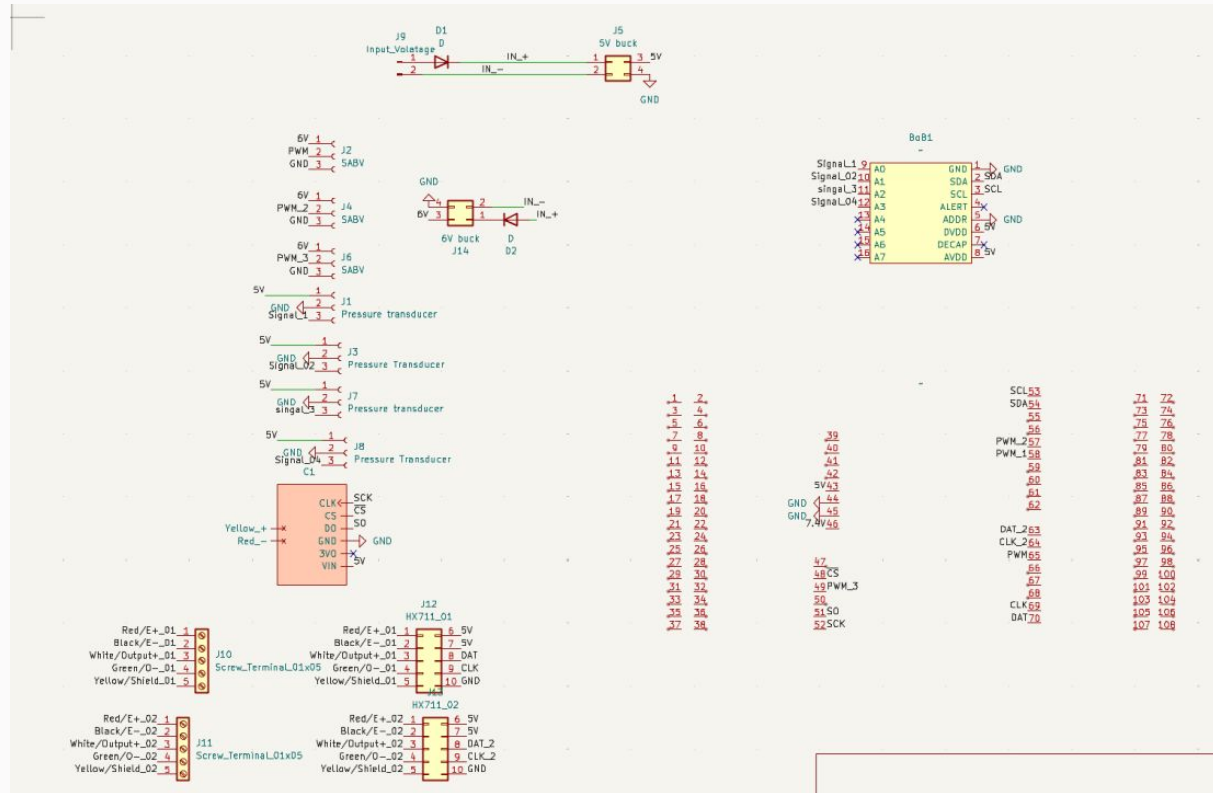
Remote Fill - Hardware

Criteria:

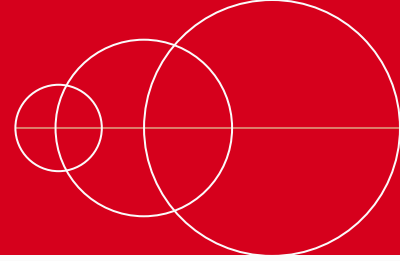
- Control remote fill system
- Monitor tank propellant mass
- Monitor saturation temperature of the tank during launch
 - Disconnect via EQD

Further Development

Test current draw from SABVs, finalize power supply circuitry, design a custom PCB



Remote Fill - Firmware

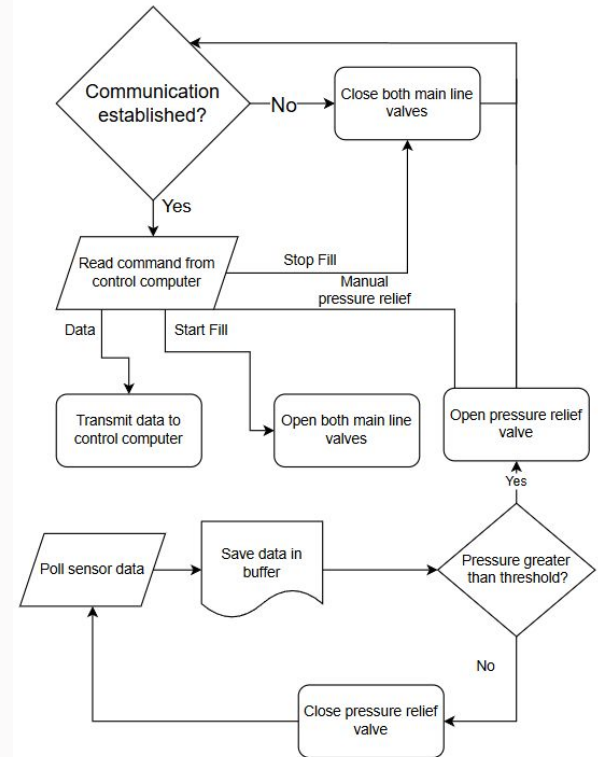


Criteria:

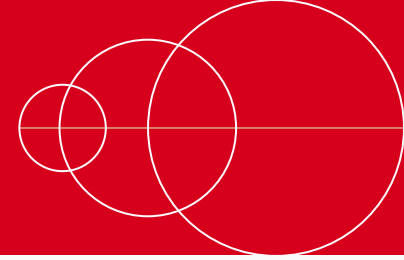
- Halt fill if communication is lost
- Initiate/halt fill on command
- Initiate pressure relief on command
- Automatically control pressure relief system
- Transmit pressure, force, and temperature readings to control computer

Further Development

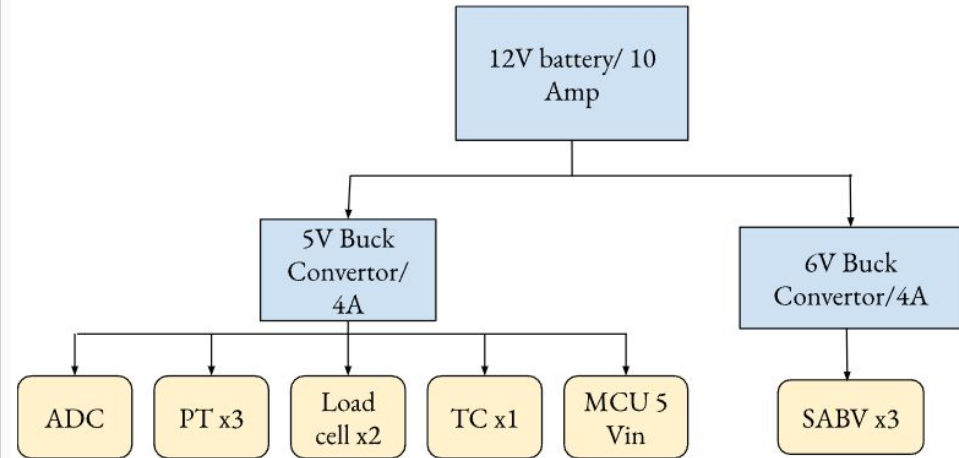
Finalize command requirements, verify readings from sensors



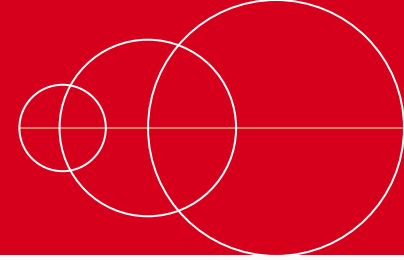
Remote Fill - Power



Component	Voltage	Current	Quantity
Microcontroller	3.3 V	<300 mA	1
Servo (40kg)	4.8V-6.8 V	<3.9A	3
ADC	5V	0.2mA	1
HX711/Loadcell	5V	10-30mA	2
MAX31855	3.3V-5V	1500 μ A	1
Pressure Transducer	5V	4-20mA	3



Communications



Overview

- 3 peripheral devices
 - DAQ, Fill, Engine Controller
- 1 master device
 - Control computer
- Queue system for commands
 - Priority for shutdown
- Parameter system for test sequences

Criteria:

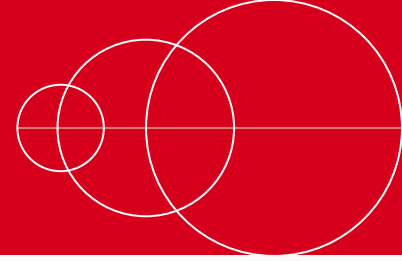
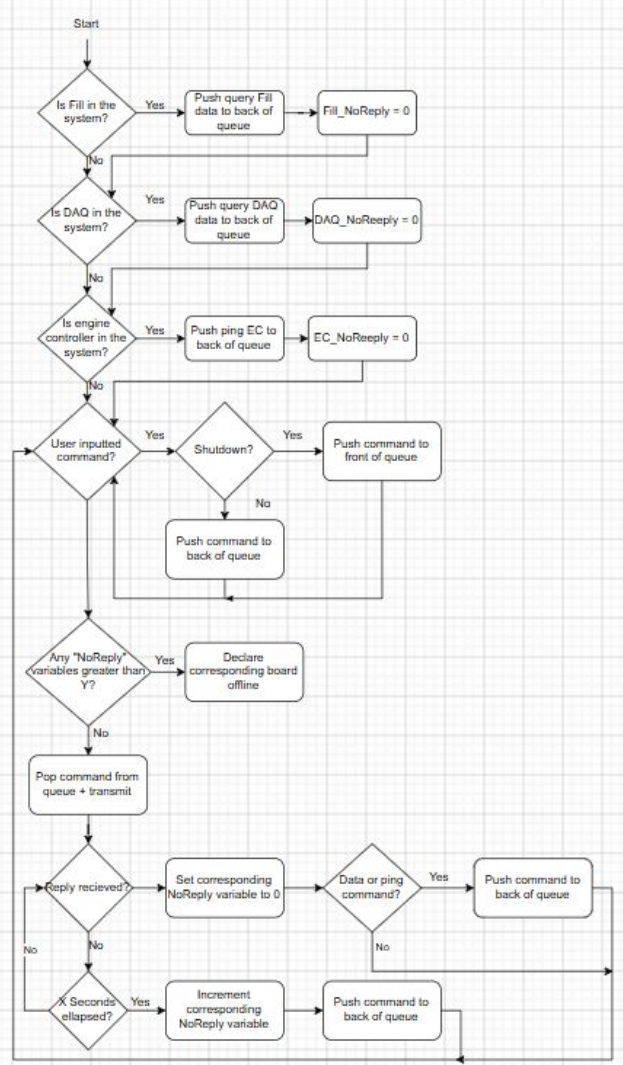
- Communicate from 2000+ feet over 900 MHz LoRa
- Constantly verify that each peripheral is online
- Error checking



Further Development

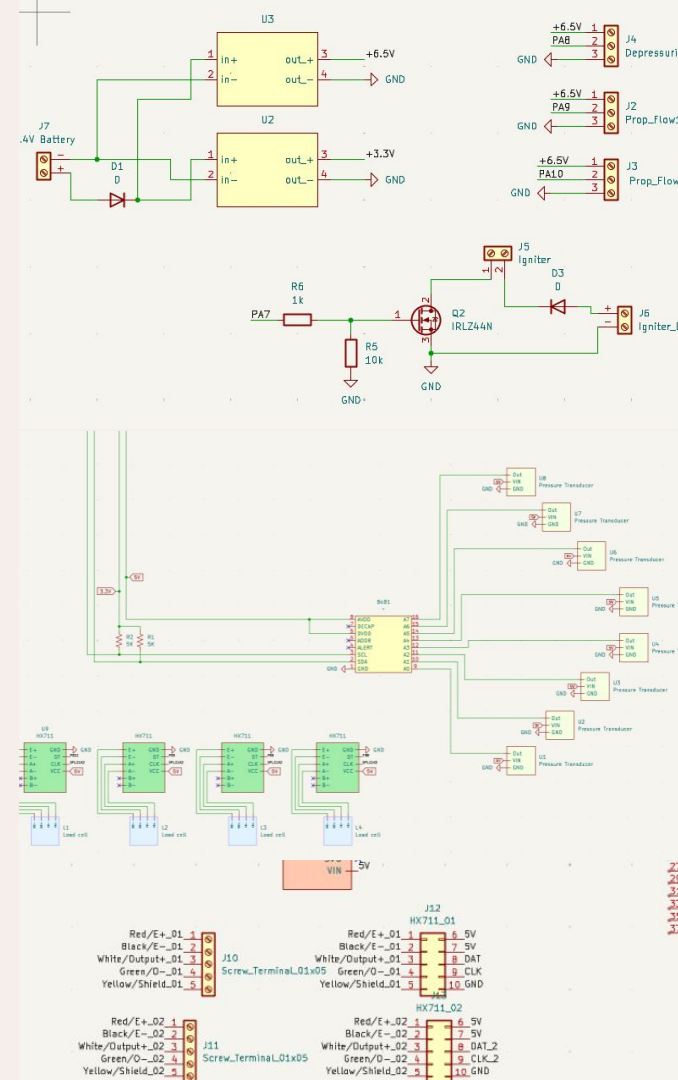
Small-scale experiments with radio, experimentally determine response time + threshold for declaring a device offline, ensure that radio can be transmitted through lower fluids bay

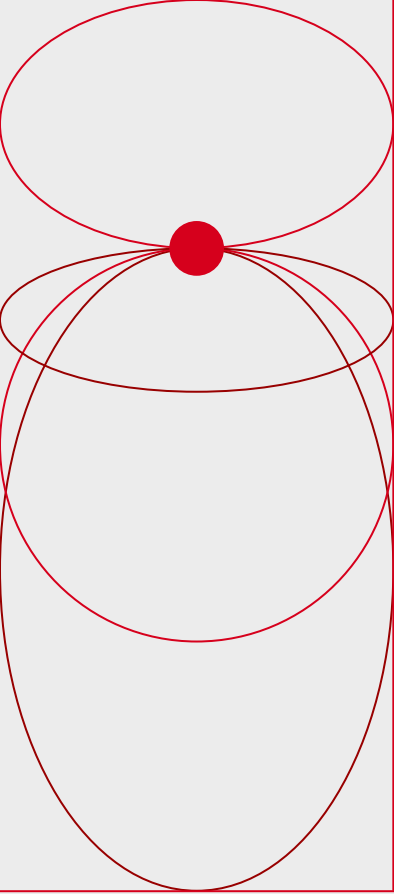
Communications



Notable Interfaces - E & S

Relevant Components	Importance (1→3)	Brief Explanation
DAQ -> Sensors	2	Data collection from sensors
Fill -> Sensors	3	Data collection from sensors
Fill -> SABVs	3	Controlling fill/ pressure relief
Engine Controller -> SABVs	3	Controlling propellant flow
Engine Controller -> Igniter	3	Igniting the rocket





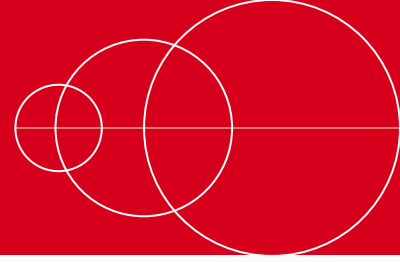
Testing, Operations, and Validation

Subsystem Engineering Review

Subsystem Lead: Roberto Ribay

Subsystem Members: Nicholas Gladu, Matthew Sweitzer, Josh de Cesare, Ryan Ing, Yeshe Dorje Yodsampa

Preliminary Test Cases

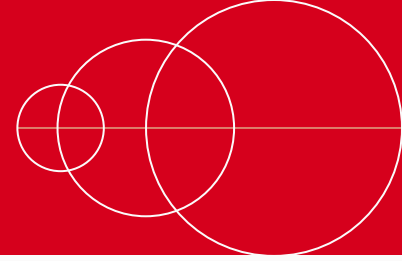


Test cases generated from requirements and selected to become associated TPMs

Planned Tests:

- Subsystem Assembly Inspections
- Nitrous Tank and Piston Hydrostatic Test
- Tank Support Structure Test
- Cold Flow Test
- Hot-Fire Test
- Integrated Static Fire Test

Failure Modes, Effects, and Criticality



Analysis

Completed FMECA of major components in accordance with NASA guidelines [GSFC-HDBK-8004]

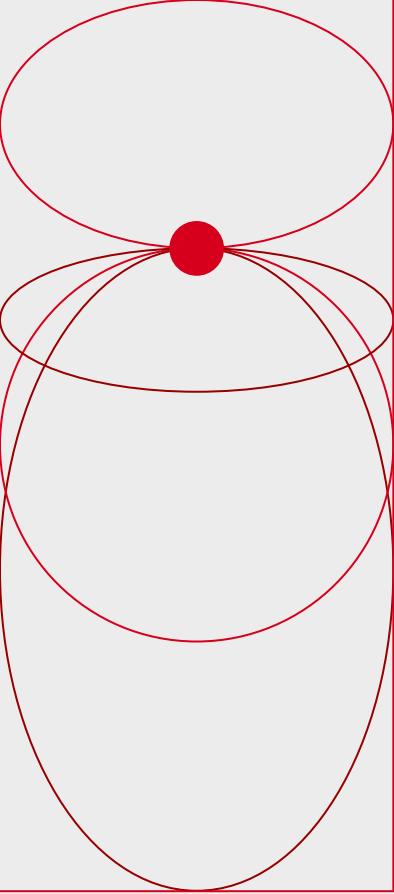
RPN Calculation Criteria:

- Likelihood
- Severity
- Detection/Prevention

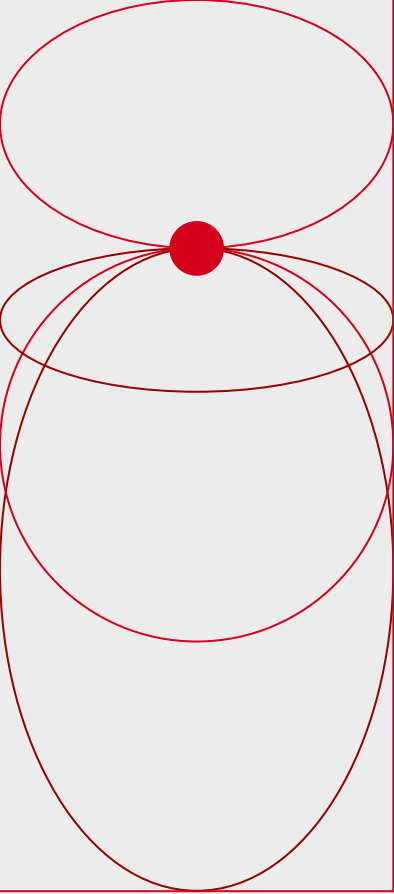
Failure Mode	Detection/Prevention/Mitigation
Thermal Shock	Proper Material Selection, Simulations, Testing,
Excessive Force	Sufficient FOS, FEA Sims, Testing
Sealing Failure	Testing, Redundant Seals

High Priority Components

Components	Relative Importance (1→3)	Component Purpose
Pintle Tip	3	Deflects propellants radially outward
Bulkhead Bolts and Bolt Holds	2	Assemble the bulkheads together
Drogue Parachute	2	Reduce velocity before main parachute deployment
Load Cell Spacer	2	Centers the engine by aligning load cells
Propellant Outer Annulus	1	Surrounds the pintle post and channels the second propellant to intersect with the first propellant



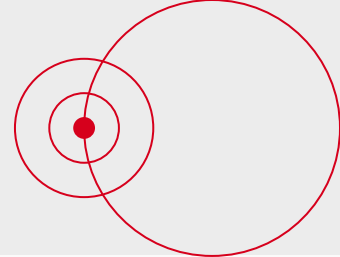
Comments & Advice



Conclusion & Evaluation

PDR Exit Criteria Status

18 Total Exit Criteria



18

CLOSED

0

ON TRACK

0

NOT ON TRACK

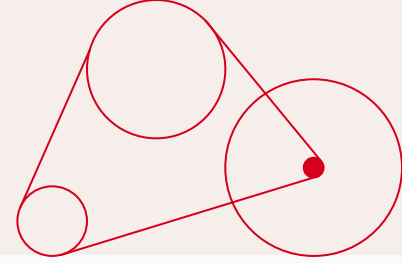
Status of Key Exit Criteria

#	Criterion	Status
1	Top-level requirements, TPMs, constraints finalized	CLOSED
2	Requirements flow-down complete and traceable	CLOSED
3	Cost, schedule, JCL are credible & within constraints	CLOSED
4	Preliminary design expected to meet requirements	CLOSED
5	System interface definitions consistent with maturity	CLOSED

Exit #9 margins closure plan: mass/power budget verification rolls up from subteams by end of April; formal baseline at CDR.

Path Forward to CDR

Phase B (PDR) to Phase C (CDR)



IMMEDIATE NEXT STEPS (2 WEEKS POST-PDR)

- Capture & disposition all PDR RFAs/RIDs in tracker
- Close out remaining PDR open items (6.16, 6.17, 9)
- Begin DMLS print slot reservation for combustion chamber
- Start long-lead valve/sensor procurement quotes
- Finalize mass/power budgets with subteam rollup

CDR DELIVERABLES — 90% DESIGN MATURITY

- Manufacturing drawings for all flight parts
- Validated FEA/CFD for pressure vessels & chamber
- Signed & frozen ICDs across all subsystems
- Complete FMECA with mitigation owners assigned
- Design freeze — changes gated via ECR/CCB

KEY MILESTONES

CDR

Oct 2026

Design freeze

Procure

Oct 2026 - Feb 2027

Long-lead items
ordered

Hot Fire & TRR

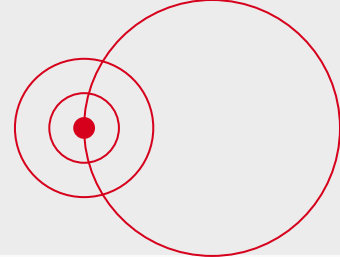
Jan - Feb 2026

Static fire campaign

FRR

Apr 2027

Flight go-ahead



REQUESTED ACTION

Does the Technical Review Board approve the Aquila program to proceed to Phase C (Critical Design Review)?

GO

Proceed to CDR



**GO WITH
ACTIONS**

Proceed with RFAs logged



NO-GO

Re-review required



PDR CRITERIA SUMMARY: 47 Closed • 1 On Track • 0 Not On Track (all with closure plans)



RPI Experimental Propulsion Initiative

Thank You

Questions & Discussion